# NANO-MAMBA U-NET: A SPATIO-TEMPORAL FRAMEWORK FOR CARDIAC MOTION ANALYSIS

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# NANO-MAMBA U-NET: A SPATIO-TEMPORAL FRAMEWORK FOR CARDIAC MOTION ANALYSIS

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**NANO-MAMBA U-NET: A SPATIO-TEMPORAL FRAMEWORK FOR CARDIAC MOTION ANALYSIS**

## ABSTRACT

Cardiac Magnetic Resonance (CMR) imaging is widely used for quantitative assessment of cardiac anatomy and function. In automated cardiac analysis, accurate segmentation of the right ventricle (RV), myocardium (MYO), and left ventricle (LV) is an important prerequisite for deriving functional measurements such as end-diastolic volume, end-systolic volume, myocardial mass, and ejection fraction. Although U-Net-style convolutional neural networks remain strong foundations for medical image segmentation, practical deployment also requires attention to model size, inference cost, and reproducible validation. This thesis investigates Nano-Mamba U-Net, a lightweight Mamba-inspired 3D U-Net architecture for cardiac MRI segmentation. The model keeps the encoder-decoder structure and skip connections of U-Net while inserting a compact sequence-processing bottleneck at the compressed latent representation. This bottleneck serializes a 3D spatial feature grid into a one-dimensional sequence, applies a local gated module inspired by Mamba-style sequence modeling, and reshapes the output back into volumetric form. The trained backbone is spatial: tensor depth denotes anatomical slices, and labelled end-diastolic (ED) and end-systolic (ES) volumes were used as independent training samples. The implementation is therefore Mamba-inspired rather than a reproduction of the original selective scan algorithm. The model was evaluated on the Automated Cardiac Diagnosis Challenge (ACDC) dataset using a deterministic patient-level validation protocol. The six-model experiment used 80 training patients and 20 validation patients, corresponding to 160 training cases and 40 validation cases. Nano-Mamba U-Net achieved 84.78% validation mean Dice Similarity Coefficient (DSC) with 1.46 million reported parameters. It exceeded the evaluated 3D U-Net baseline at 80.83% DSC, but SegResNet16 achieved the highest DSC of 86.70% and the convolutional-bottleneck control (No-Mamba) achieved 85.64%. The principal architectural result is therefore a compact accuracy–efficiency trade-off rather than best Dice. A separate post-hoc full-cine analysis applied the selected spatial Nano-Mamba checkpoint to all 550 frames from the same 20-patient validation cohort. Each 3D frame was preprocessed independently, and fixed circular probability fusion, $0.25P_{t-1} + 0.50P_t + 0.25P_{t+1}$, was compared with frame-wise inference. The analysis reconstructed native-grid masks, ventricular volume trajectories, ED/ES timing, ejection fraction (EF), LV-centroid displacement, and a global myocardial radial-distance surrogate. Temporal fusion produced nearly unchanged resized-grid endpoint DSC (84.794% versus 84.779%) and native-grid endpoint DSC (77.923% versus 77.919%); descriptive EF mean absolute error was 3.329 versus 3.514 percentage points. Paired 95% bootstrap intervals for these differences included zero. Normalized LV-curve second difference decreased by 0.00113, with a paired interval of $[-0.00195, -0.00051]$ and smoother curves in 19 of 20 patients. Thus, the framework supports segmentation-derived full-cine functional and global motion analysis, but it is not a learned temporal Mamba model, optical flow, dense deformation, or regional strain estimation. Intermediate frames lack manual masks, and the validation cohort was also used for checkpoint selection; independent testing, multi-seed training, external validation, and learned temporal or dense-motion modeling remain future work.

**Keywords:** Cardiac MRI Segmentation, Cine MRI, Spatio-Temporal Analysis, Mamba-Inspired Architecture, Accuracy-Efficiency Trade-off.

**NANO-MAMBA U-NET: RANGKA KERJA RUANG-MASA UNTUK ANALISIS PERGERAKAN JANTUNG**

## ABSTRAK

Pengimejan Resonans Magnetik Jantung (CMR) digunakan secara meluas untuk penilaian kuantitatif anatomi dan fungsi jantung. Dalam analisis jantung automatik, segmentasi tepat bagi ventrikel kanan (RV), miokardium (MYO), dan ventrikel kiri (LV) penting sebelum ukuran fungsi seperti isipadu akhir diastol, isipadu akhir sistol, jisim miokardium, dan pecahan ejeksi boleh diperoleh. Tesis ini mengkaji Nano-Mamba U-Net, iaitu seni bina 3D U-Net ringan yang diinspirasikan oleh Mamba untuk segmentasi MRI jantung. Model mengekalkan struktur pengekod-penyahkod dan sambungan langkau U-Net sambil memasukkan bottleneck pemprosesan jujukan padat pada perwakilan laten termampat. Bottleneck menyusun grid ciri spatial 3D kepada jujukan satu dimensi, menggunakan modul berpagar setempat yang diinspirasikan oleh Mamba, dan membentuk semula output kepada bentuk volumetrik. Rangkaian terlatih bersifat spatial: dimensi kedalaman mewakili hirisan anatomi, manakala isipadu akhir diastol (ED) dan akhir sistol (ES) digunakan sebagai sampel latihan berasingan. Oleh itu, pelaksanaan ini diinspirasikan oleh Mamba dan bukannya pelaksanaan algoritma selective scan asal. Model dinilai menggunakan set data Automated Cardiac Diagnosis Challenge (ACDC) dengan protokol pengesahan berasaskan pesakit. Eksperimen enam model menggunakan 80 pesakit untuk latihan dan 20 pesakit untuk pengesahan, bersamaan 160 kes latihan dan 40 kes pengesahan. Nano-Mamba U-Net mencapai min Dice Similarity Coefficient (DSC) pengesahan 84.78% dengan 1.46 juta parameter yang dilaporkan. Model ini mengatasi garis dasar 3D U-Net pada 80.83% DSC, tetapi SegResNet16 mencapai DSC tertinggi 86.70% dan kawalan bottleneck konvolusi (No-Mamba) mencapai 85.64%. Oleh itu, hasil seni

bina utama ialah keseimbangan ketepatan–kecekapan yang padat, bukannya DSC terbaik. Analisis cine lengkap pasca-hoc yang berasingan kemudian menggunakan checkpoint Nano-Mamba spatial terpilih pada kesemua 550 bingkai daripada kohort pengesahan 20 pesakit yang sama. Setiap bingkai 3D dipraproses secara berasingan, dan gabungan kebarangkalian bulat tetap, $0.25P_{t-1} + 0.50P_t + 0.25P_{t+1}$, dibandingkan dengan inferens setiap bingkai. Analisis membina semula topeng pada grid asal, trajektori isipadu ventrikel, masa ED/ES, pecahan ejeksi (EF), anjakan sentroid LV, dan ukuran pengganti global jarak jejari miokardium. Gabungan temporal menghasilkan DSC titik akhir pada grid ubah saiz yang hampir tidak berubah (84.794% berbanding 84.779%) dan DSC grid asal (77.923% berbanding 77.919%); ralat mutlak purata EF deskriptif ialah 3.329 berbanding 3.514 mata peratusan. Selang bootstrap berpasangan 95% bagi perbezaan ini merangkumi sifar. Perbezaan kedua ternormalisasi trajektori LV berkurang sebanyak 0.00113, dengan selang $[-0.00195, -0.00051]$ dan trajektori lebih licin untuk 19 daripada 20 pesakit. Oleh itu, rangka kerja menyokong analisis fungsi dan pergerakan global sepanjang kitaran cine yang diterbitkan daripada segmentasi, tetapi bukan model Mamba temporal terlatih, aliran optik, ubah bentuk padat, atau anggaran regangan setempat. Bingkai perantaraan tidak mempunyai topeng manual dan kohort pengesahan turut digunakan untuk pemilihan checkpoint; ujian bebas, latihan berbilang benih, pengesahan luaran, serta pemodelan temporal terlatih atau gerakan padat kekal sebagai kerja masa hadapan.

**Kata Kunci:** Segmentasi MRI Jantung, MRI Cine, Analisis Ruang-Masa, Seni Bina Diinspirasikan Mamba, Keseimbangan Ketepatan-Kecekapan.

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**TABLE OF CONTENTS**

# LIST OF FIGURES

## LIST OF TABLES

## LIST OF SYMBOLS AND ABBREVIATIONS

| | | |
|---|---|---|
| $D$ | : | anatomical slice-depth dimension; not cardiac time. |
| $P_t$ | : | class-probability tensor at cine phase $t$. |
| $V_{LV,t}$ | : | predicted left-ventricular volume at cine phase $t$. |
| $\Delta$ | : | paired difference between two executed methods. |
| ACDC | : | Automated Cardiac Diagnosis Challenge. |
| AMP | : | automatic mixed precision. |
| CI | : | confidence interval. |
| CMR | : | cardiac magnetic resonance. |
| DSC | : | Dice Similarity Coefficient. |
| ED | : | end diastole. |
| EDV | : | end-diastolic volume. |
| EF | : | ejection fraction. |
| ES | : | end systole. |
| ESV | : | end-systolic volume. |
| FPS | : | frames per second. |
| LV | : | left ventricle. |
| MAE | : | mean absolute error. |
| MRI | : | magnetic resonance imaging. |
| MYO | : | myocardium. |
| NIfTI | : | Neuroimaging Informatics Technology Initiative file format. |
| RV | : | right ventricle. |
| SSM | : | state space model. |
| SV | : | stroke volume. |

# CHAPTER 1: INTRODUCTION

## 1.1 Background of the Study

### 1.1.1 Clinical Significance of Cardiovascular Diseases

Cardiovascular diseases (CVDs) remain one of the most important global causes of mortality. According to the World Health Organization (WHO), CVDs caused an estimated 19.8 million deaths in 2022, corresponding to approximately 32% of all global deaths (World Health Organization, 2025). This figure is not used in this thesis as a direct claim about the performance of any computational model. Rather, it establishes the clinical context in which efficient and reproducible cardiac image analysis tools are valuable. When the global disease burden is high, even small improvements in the reliability or scalability of cardiac assessment workflows can have practical importance.

Cardiac Magnetic Resonance (CMR) imaging is widely used for non-invasive assessment of cardiac anatomy and function (Bernard et al., 2018). From CMR images, clinicians can derive functional indices such as Ejection Fraction (EF), End-Diastolic Volume (EDV), End-Systolic Volume (ESV), and Myocardial Mass. These measurements depend on accurate delineation of the Right Ventricle (RV), Myocardium (MYO), and Left Ventricle (LV). Therefore, automated segmentation of these structures is a central step in computer-assisted cardiac analysis. This project first evaluates spatial ED/ES segmentation and then applies the selected Nano-Mamba checkpoint across complete cine sequences to derive ventricular volume trajectories, EF, phase timing, and global segmentation-derived motion surrogates.

The clinical relevance of this task comes from the relationship between segmentation masks and quantitative cardiac indices. If $V_{ED}$ denotes the ventricular volume at end-diastole and $V_{ES}$ denotes the volume at end-systole, ejection fraction can be expressed as:

$$EF = \frac{V_{ED} - V_{ES}}{V_{ED}} \times 100\%. \tag{1.1}$$

This equation shows why segmentation errors matter. A small boundary error can change the estimated cavity volume, and the resulting volume error can propagate into EF. In the completed full-cine analysis, the same relationship is evaluated at the manually labelled ED/ES endpoints and used to construct a complete segmentation-derived LV-volume trajectory.

### 1.1.2 The Bottleneck of Manual Delineation and Traditional Methods

Manual delineation of cardiac structures requires expert knowledge and substantial effort. A cine CMR study may contain many slices and cardiac phases. The ACDC challenge paper emphasizes that accurate segmentation of RV, MYO, and LV is essential for cardiac diagnosis and that manual annotation can be affected by inter-observer variability (Bernard et al., 2018). Instead of relying on an unsupported fixed time estimate, this thesis treats manual annotation as a practical bottleneck because it is labor-intensive, repetitive, and difficult to scale across large datasets.

Manual segmentation is also not purely mechanical. The myocardium has an annular shape and may be difficult to separate from papillary muscles. The RV has a thin wall and variable geometry. The basal and apical slices often contain ambiguous anatomy because the visible structures gradually appear or disappear across the slice direction. These difficulties mean that even expert annotations may vary. Automated segmentation is therefore attractive not because it removes the need for clinical expertise, but because it can provide a consistent computational starting point for measurement and review.

Traditional image processing methods, including thresholding, active contours, level sets, and graph-cut algorithms, were historically applied to medical image segmentation.

However, these methods rely heavily on hand-crafted features, manual initialization, or assumptions about image intensity distributions. In CMR images, boundary ambiguity, partial volume effects, papillary muscles, and scanner-dependent intensity variations make fully automated segmentation challenging (Chen et al., 2020). These limitations motivated the transition toward deep learning methods that learn task-specific representations from annotated data.

### 1.1.3 Deep Learning for Cardiac Segmentation

Deep learning shifted medical image segmentation from hand-crafted feature engineering toward data-driven representation learning. Fully Convolutional Networks (FCNs) introduced dense pixel-wise prediction for semantic segmentation (Long et al., 2015), while U-Net became a standard architecture for biomedical image segmentation due to its encoder-decoder design and skip connections (Ronneberger et al., 2015). For volumetric medical images, 3D U-Net extended this idea to three-dimensional convolutional processing (Cicek et al., 2016). These architectures are important foundations for this thesis, and this project does not claim them as original contributions.

Despite their success, convolutional networks still rely on local kernels. A $3 \times 3 \times 3$ convolution extracts information from a local voxel neighbourhood. Stacking several convolutional layers increases the effective receptive field, but the operation remains locally constructed. In cardiac MRI, local detail and wider anatomical context are both important. The model must distinguish the myocardium from the blood pool at local boundaries while also maintaining enough contextual awareness to understand the relative positions of RV, MYO, and LV.

Transformer-based architectures address long-range interactions through self-attention (Vaswani et al., 2017; Dosovitskiy et al., 2021). Medical segmentation models such as UNETR and Swin UNETR demonstrate the usefulness of global or window-based token interactions in volumetric segmentation tasks (Hatamizadeh et al., 2022b,a). However, self-attention can increase memory and computational requirements, which motivates research into more efficient sequence modeling alternatives. This thesis does not argue that Transformers are unsuitable for medical imaging. Instead, it investigates whether a lighter sequence-inspired bottleneck can be useful when the hardware environment is constrained.

### 1.1.4 Motivation for a Mamba-Inspired Bottleneck

Selective State Space Models (SSMs), including Mamba, have recently attracted attention because they can model long sequences with linear-time complexity (Gu and Dao, 2023). This thesis investigates whether a lightweight Mamba-inspired sequence bottleneck can be integrated into a 3D U-Net-style architecture for cardiac MRI segmentation. The implemented model serializes compressed 3D volumetric features into a one-dimensional token sequence, applies a lightweight gated sequence module inspired by Mamba, and reshapes the result back into volumetric form.

The motivation is based on the accuracy-efficiency trade-off. A model with the highest Dice score is not necessarily the most suitable model for all environments if it requires more memory, more parameters, or longer inference time. Conversely, a very small model is not useful if it fails to segment clinically relevant structures. This thesis therefore evaluates Nano-Mamba U-Net not only by segmentation accuracy but also by parameter count and inference speed. The research question is whether a compact sequence-inspired bottleneck can produce competitive validation performance while maintaining a small model size.

The completed empirical design has two linked stages. In the historical six-model experiment, labelled ED and ES volumes are processed as independent 3D cases and tensor depth contains anatomical slices rather than time. In a separate post-hoc stage, the selected spatial Nano-Mamba checkpoint is applied to every frame of each validation cine. Fixed adjacent-frame probability fusion is then evaluated against frame-wise inference, and the resulting native-grid masks are used to compute ventricular volume trajectories, ED/ES timing, EF, LV-centroid displacement, and a global myocardial radial-distance surrogate. The project therefore completes a segmentation-derived spatio-temporal analysis, while stopping short of claiming a learned temporal Mamba network, optical flow, dense deformation, or regional strain.

## 1.2 Scope of the Study

The scope of this research is defined as follows:

- **Dataset Boundary:** The empirical evaluation uses the Automated Cardiac Diagnosis Challenge (ACDC) dataset (Bernard et al., 2018). The segmentation targets are RV, MYO, and LV.
- **Task Boundary:** The trained backbone performs four-class 3D cardiac MRI segmentation and treats ED/ES cases independently. A separate full-cine workflow evaluates frame-wise inference and fixed temporal probability fusion across 550 validation frames, then derives global volume, phase, EF, centroid-displacement, and myocardial-radius trajectories. Slice depth is never treated as time.
- **Implementation Boundary:** The proposed Nano-Mamba U-Net uses a Mamba-inspired bottleneck implemented in PyTorch. It should not be interpreted as a full reproduction of the original hardware-aware selective scan implementation.
- **Evaluation Boundary:** The main reported results are based on a deterministic patient-level 80/20 validation split of the ACDC training cohort. The study does not claim performance on the hidden official ACDC test set.
- **Hardware Boundary:** Training, validation, and computational profiling were

conducted on a single consumer-grade NVIDIA RTX 4060 GPU with 8GB VRAM. Reported FPS is specific to the implemented batch-one benchmark and is not treated as a universal property of an architecture.

These boundaries are deliberately stated to avoid over-claiming. The goal of the thesis is to produce a rigorous and reproducible project-level investigation, not to claim clinical deployment readiness or universal superiority over all existing cardiac segmentation methods.

## 1.3 Organization of the Thesis

This thesis is organized into eight chapters. Chapter 1 introduces the clinical and computational motivation. Chapter 2 defines the research problem. Chapter 3 states the objectives and research questions. Chapter 4 reviews relevant work in cardiac image segmentation, CNNs, attention models, residual networks, and state space models. Chapter 5 describes the dataset, preprocessing pipeline, architecture, training protocol, and evaluation method. Chapter 6 reports the validation results. Chapter 7 discusses the findings and limitations. Chapter 8 concludes the thesis and outlines future work.

# CHAPTER 2: RESEARCH PROBLEM

Automated cardiac MRI segmentation is not only an accuracy problem. A clinically useful model must also be computationally practical and robust across different cardiac anatomies. This thesis focuses on the gap between accurate volumetric segmentation and lightweight implementation. The problem is framed around three connected issues: local context modeling, computational cost, and anatomical variability.

## 2.1 Local Context Modeling in Lightweight 3D CNNs

3D CNNs are effective for volumetric segmentation because they process spatial neighborhoods directly. However, a standard $3 \times 3 \times 3$ convolution only aggregates local voxel information at each layer. Although deeper networks can enlarge the effective receptive field, lightweight models with limited depth and channel capacity may still struggle to capture broader contextual relationships across the heart. This limitation is important in CMR segmentation because RV, MYO, and LV boundaries can be ambiguous at basal and apical slices.

The issue is not that convolution is ineffective. Convolution is one of the reasons U-Net-style models are successful in medical segmentation. The problem is that a lightweight model must compress information aggressively to remain small. This thesis therefore tests whether a compact gated transformation along a raster-serialized latent representation provides a useful engineering trade-off. The executed block only mixes locally with a kernel of three along that sequence; it does not provide selective state propagation or direct global all-token interaction. This is the specific architectural motivation and boundary for inserting the block at the bottleneck rather than replacing the entire U-Net with a large attention model.

## 2.2 Computational Cost of Larger Segmentation Models

Stronger segmentation architectures often increase depth, width, attention operations, or residual blocks to improve representation capacity. This can improve accuracy but also increases parameters, memory usage, and inference cost. For resource-constrained clinical, educational, or prototype environments, a model with lower parameter count and acceptable accuracy may be preferable to a heavier model with marginally higher Dice scores. Therefore, the central problem is not simply to maximize Dice, but to study the trade-off between segmentation accuracy and computational efficiency.

This thesis uses parameter count and inference speed as supporting metrics because Dice alone does not describe the full engineering value of a model. Here, *lightweight* means the lowest reported trainable-parameter count together with competitive batch-one throughput on the recorded hardware; it does not mean leadership in every speed, memory, or accuracy metric. A model with 1.46M parameters and competitive Dice may be useful even if it is not the highest-accuracy model. Conversely, if a lightweight model achieves lower Dice without meaningful savings, it would not be justified. The final evaluation therefore compares Nano-Mamba U-Net against baselines and ablation models using the same split and training protocol.

## 2.3 Pathological and Anatomical Variability

The ACDC dataset contains normal subjects and multiple cardiac pathology groups, including dilated cardiomyopathy, hypertrophic cardiomyopathy, myocardial infarction, and abnormal right ventricle cases (Bernard et al., 2018). These cases introduce variation in ventricular size, myocardial thickness, and boundary appearance. A segmentation model must therefore handle both normal cardiac geometry and pathological deformation.

Anatomical variability is particularly important for the RV and MYO. The RV has a complex shape and thinner wall, which makes its segmentation more sensitive to boundary ambiguity. The MYO is a narrow structure, so small absolute boundary errors can cause a noticeable Dice reduction. This explains why class-specific Dice is reported in the results chapter instead of reporting only one aggregate number. The evaluation must show whether the model performs consistently across RV, MYO, and LV.

## 2.4 From Endpoint Segmentation to Cardiac-Cycle Analysis

Labelled ED and ES volumes are sufficient for a conventional segmentation benchmark, but they do not show how predictions evolve over the intervening cardiac phases. Applying a spatial model independently to every cine frame can produce a complete segmentation trajectory, yet adjacent predictions may fluctuate because the network was not trained with temporal context. The resulting research problem is therefore to connect the validated spatial checkpoint to a transparent temporal analysis without misrepresenting it as a learned four-dimensional model. This thesis addresses that gap with fixed circular probability fusion and explicitly evaluates endpoint overlap, EF, phase timing, curve smoothness, and global motion surrogates. Manual endpoint labels provide partial validation, while the absence of intermediate-frame labels remains a stated limitation.

## 2.5 Research Gap

Existing segmentation backbones such as 3D U-Net, Attention U-Net, and SegResNet provide strong baselines, but they represent different points on the accuracy-efficiency spectrum. The first research gap addressed in this thesis is whether a compact Mamba-inspired bottleneck can provide competitive validation performance with fewer parameters than conventional or residual alternatives, while remaining feasible on a consumer GPU. The second is whether the selected spatial checkpoint can support a reproducible full-cine, segmentation-derived function and global motion analysis, and whether a fixed temporal fusion produces a measurable change relative to independent frame-wise inference.

The gap is not a claim that previous methods are inadequate in all settings. SegResNet, for example, remains a strong residual convolutional baseline. The gap is narrower and more appropriate for this project: there is value in testing a compact sequence-inspired bottleneck under a rigorous patient-level validation protocol, especially when earlier exploratory scripts may overestimate performance by evaluating on non-held-out data. This thesis therefore emphasizes reproducibility and honest comparison as much as architectural novelty.

# CHAPTER 3: RESEARCH OBJECTIVES

The aim of this research is to design, implement, and evaluate a lightweight Mamba-inspired U-Net architecture for cardiac MRI segmentation and to determine how its segmentations can support a bounded full-cine cardiac motion and function analysis. The study emphasizes empirical validation, parameter efficiency, transparent comparison with established segmentation baselines, and explicit separation between spatial learning and post-hoc temporal processing. The objectives connect directly to the experiments reported in Chapter 6.

## 3.1 Specific Objectives

1. To implement a Nano-Mamba U-Net architecture that inserts a lightweight Mamba-inspired sequence bottleneck into a 3D U-Net-style encoder-decoder network.
2. To evaluate the segmentation performance of the proposed model on RV, MYO, and LV using Dice Similarity Coefficient on a deterministic patient-level validation split of the ACDC training cohort.
3. To compare the proposed model with 3D U-Net, Attention U-Net, SegResNet, and ablation variants using the same split, preprocessing, epoch budget, optimizer family, and validation protocol, with model-specific batch sizes disclosed as a limitation.
4. To quantify computational efficiency using reported trainable-parameter count and batch-one inference speed.
5. To apply the selected Nano-Mamba checkpoint to complete validation cine sequences, derive segmentation-based cardiac volume, phase, EF, and global motion trajectories, and quantify what fixed adjacent-frame probability fusion changes relative to independent frame-wise inference.

The first objective is implemented through the model architecture described in Chapter

5. The second and third objectives are addressed through the six-model quantitative validation table in Chapter 6. The fourth objective is addressed by reporting trainable parameters and inference speed together with Dice. The fifth objective is addressed by the independently audited 20-patient, 550-frame full-cine analysis. This mapping ensures that every stated objective is answered by an implemented experiment.

## 3.2 Research Questions

1. How can compressed 3D cardiac MRI feature maps be serialized and processed by a Mamba-inspired bottleneck within a U-Net-style segmentation architecture?
2. Does the proposed Nano-Mamba U-Net achieve competitive held-out validation Dice compared with 3D U-Net, Attention U-Net, SegResNet, and ablation variants?
3. What accuracy-efficiency trade-off is observed when comparing Nano-Mamba U-Net with heavier or non-Mamba alternatives in terms of Dice score, parameter count, and inference speed?
4. How can the selected spatial segmentation checkpoint support full-cine cardiac function and global motion analysis, and what measurable effect does fixed adjacent-frame probability fusion have relative to frame-wise inference?

These research questions are intentionally phrased in a measurable way. The first question is answered by architecture implementation rather than by a numerical score alone. The second question is answered by held-out validation Dice. The third question is answered by jointly interpreting Dice, parameter count, and speed. The fourth is answered by complete cine coverage, endpoint validation, functional and motion-derived quantities, and paired temporal-fusion comparisons. This structure avoids unsupported claims such as absolute superiority or dense motion tracking and keeps the thesis aligned with the actual evidence.

# CHAPTER 4: LITERATURE REVIEW

This chapter reviews the scientific and technical literature that supports the design choices of this thesis. The purpose is not to claim that the proposed model replaces all existing cardiac segmentation methods. Instead, the review identifies the specific research context in which a lightweight Mamba-inspired U-Net is meaningful: automated cardiac MRI segmentation requires accurate anatomical delineation, but practical models must also consider parameter count, memory usage, and reproducible validation.

## 4.1 Cardiac MRI Segmentation as a Clinical Computing Problem

Cardiac Magnetic Resonance (CMR) imaging is widely used for non-invasive assessment of cardiac morphology and function. In the ACDC benchmark, the segmentation targets are the Right Ventricle (RV), Myocardium (MYO), and Left Ventricle (LV), which are clinically important because they support the computation of ventricular volumes, myocardial mass, and ejection fraction (Bernard et al., 2018). These quantities are not obtained directly from a neural network classifier; they require spatial masks from which anatomical volumes can be computed. Therefore, segmentation quality is central to the reliability of downstream cardiac function analysis.

CMR segmentation is challenging because the difficulty is not uniform across all structures. LV segmentation is often more stable because the LV cavity is relatively compact and has a clearer blood-pool boundary. In contrast, the myocardium is a thin ring-like structure whose boundaries can be affected by partial volume effects, papillary muscles, and signal ambiguity. The RV is also difficult because it has a more variable crescent-like shape and thinner wall. These anatomical observations explain why a single aggregate Dice score is insufficient; class-specific performance should also be reported when evaluating cardiac segmentation models.

The ACDC dataset is useful for this thesis because it includes multiple cardiac conditions, including normal subjects and pathological groups such as dilated cardiomyopathy, hypertrophic cardiomyopathy, myocardial infarction, and abnormal right ventricle cases (Bernard et al., 2018). This heterogeneity makes the benchmark more relevant than a dataset containing only healthy cardiac anatomy. Nevertheless, the ACDC training cohort remains a limited dataset, so patient-level separation between training and validation is essential. If frames from the same patient appear in both training and evaluation, the reported performance may reflect memorization of patient-specific anatomy rather than generalization.

## 4.2 Traditional Medical Image Segmentation Methods

Before deep learning became dominant, medical image segmentation commonly relied on intensity thresholding, deformable models, graph-based optimization, and level-set methods. These approaches remain important historically because they clarify why learning-based segmentation became attractive.

A simple thresholding method assumes that the target object can be separated from the background by intensity. Otsu's method, for example, selects a threshold $T$ by maximizing the between-class variance (Otsu, 1979). If the two classes have probabilities $\omega_0(T)$ and $\omega_1(T)$, and means $\mu_0(T)$ and $\mu_1(T)$, the objective can be written as:

$$\sigma_B^2(T) = \omega_0(T)\omega_1(T)\,[\mu_0(T) - \mu_1(T)]^2 . \tag{4.1}$$

This formulation is elegant because it does not require annotated training data. However, it is not sufficient for cardiac MRI segmentation. MRI intensities are not standardized in the same way as CT Hounsfield Units, and myocardium, blood pool, and surrounding tissues may overlap in intensity. As a result, a global threshold cannot reliably separate RV, MYO, and LV across patients.

Active contour models, introduced by Kass et al. (1988), formulate segmentation as an energy minimization problem. A contour evolves under internal smoothness forces and external image forces. A simplified energy functional can be written as:

$$E_{snake} = \int_0^1 \left[ \frac{1}{2}\alpha |v'(s)|^2 + \frac{1}{2}\beta |v''(s)|^2 + E_{image}(v(s)) \right] ds. \tag{4.2}$$

Here, the first two terms regulate contour continuity and smoothness, while $E_{image}$ attracts the contour to image features such as edges. This is useful when boundaries are clear, but cardiac MRI boundaries can be weak or ambiguous at basal and apical slices. Moreover, deformable models often require careful initialization, which limits their practicality for fully automated pipelines.

Level-set methods represent the contour implicitly as the zero level of a higher-dimensional function $\phi$ (Osher and Sethian, 1988). A typical evolution equation is:

$$\frac{\partial \phi}{\partial t} + F|\nabla \phi| = 0. \tag{4.3}$$

The implicit representation allows topology changes more naturally than parametric snakes. However, iterative numerical optimization and sensitivity to image forces remain practical limitations. For this reason, traditional methods are useful background, but they do not remove the need for robust data-driven segmentation in heterogeneous CMR datasets (Chen et al., 2020).

## 4.3 Deep Learning Segmentation Architectures

### 4.3.1 Fully Convolutional Networks and U-Net

Fully Convolutional Networks (FCNs) showed that classification-style neural networks could be reformulated for dense pixel-wise prediction by replacing fully connected layers with convolutional operations (Long et al., 2015). This idea was important because segmentation requires a prediction for every pixel or voxel rather than one image-level label.

U-Net became one of the most influential architectures in biomedical image segmentation (Ronneberger et al., 2015). Its main contribution is the encoder-decoder structure with skip connections. The encoder gradually reduces spatial resolution to learn semantic features, while the decoder restores resolution to produce a segmentation mask. Skip connections transfer high-resolution spatial features from the encoder to the decoder, which helps recover fine boundaries. If $\mathbf{X}_{enc}^{(l)}$ is an encoder feature map and $\mathbf{X}_{dec}^{(l+1)}$ is a decoder feature map from a coarser level, a simplified skip-connection operation can be written as:

$$\mathbf{X}_{\text{dec}}^{(l)} = \text{Conv}\left(\text{Concat}\left[\mathbf{X}_{\text{enc}}^{(l)}, \text{Upsample}\left(\mathbf{X}_{\text{dec}}^{(l+1)}\right)\right]\right), \tag{4.4}$$

where Upsample restores spatial resolution, Concat joins feature channels, and Conv performs convolutional refinement. Writing the operations by name avoids any ambiguity about the order: up-sampling occurs before channel concatenation, and convolution follows the concatenation. This equation explains why U-Net is suitable for medical segmentation: the decoder does not reconstruct boundaries only from compressed semantic features; it also receives high-resolution information from earlier encoder layers.

For volumetric data, 3D U-Net extends the same principle from 2D images to 3D tensors (Cicek et al., 2016). V-Net also used volumetric convolution and Dice-based optimization for 3D medical segmentation (Milletari et al., 2016). These works are directly relevant to this thesis because the proposed Nano-Mamba U-Net uses a 3D U-Net-style encoder-decoder as its architectural base. The contribution of this thesis is not the invention of the U-Net framework itself. Rather, the project studies whether the bottleneck of such a framework can be modified into a compact sequence-processing module.

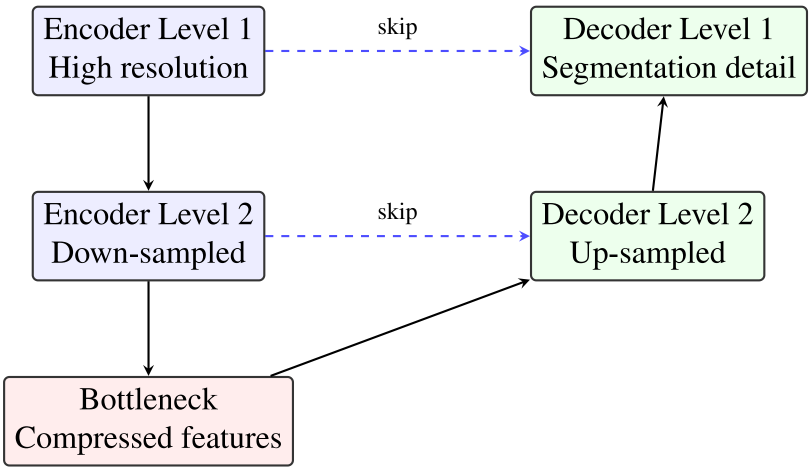


**Figure 4.1: Conceptual encoder-decoder structure of U-Net-style segmentation models. Encoder blocks learn increasingly abstract features, decoder blocks recover spatial resolution, and skip connections preserve boundary information.**

### 4.3.2 Attention U-Net

Attention U-Net introduced attention gates to suppress irrelevant encoder features before they are passed through skip connections (Oktay et al., 2018). This is relevant in medical images because large regions of an image may contain background structures unrelated to the target anatomy. In a simplified attention gate, an encoder feature $x$ and a decoder gating signal $g$ are projected into a shared representation and transformed into an attention coefficient:

$$\alpha = \mathrm{sigmoid}\Big(\psi^{\mathsf{T}}\delta(W_x x + W_g g + b)\Big). \tag{4.5}$$

The filtered feature can then be represented as:

$$\hat{x}_i = \alpha_i x_i, \tag{4.6}$$

where index $i$ runs over the spatial feature elements. Thus, sigmoid maps the attention score to a coefficient, $\delta$ is a nonlinear activation, and $\alpha_i x_i$ is explicit element-wise multiplication. The key idea is not that attention automatically guarantees better segmentation, but that it provides a mechanism for feature selection. In the final experiment of this thesis, Attention U-Net did not outperform the other models, which illustrates why empirical comparison under the same split is necessary.

### 4.3.3 Residual Networks and SegResNet

Residual learning was introduced to make deeper networks easier to optimize (He et al., 2016). Instead of asking stacked layers to learn a full mapping $H(x)$, a residual block learns a residual function $F(x)$ and adds the input back to the output:

$$y = F(x) + x. \tag{4.7}$$

This identity shortcut improves gradient flow and can make deeper convolutional models more trainable. SegResNet-style medical segmentation architectures use residual blocks in an encoder-decoder setting (Myronenko, 2019). In this thesis, the MONAI SegResNet configuration with 16 initial filters serves as a residual convolutional alternative to the proposed Nano-Mamba U-Net. The final results show that SegResNet16 achieved the highest mean Dice, which must be acknowledged rather than hidden.

## 4.4 Transformers and Sequence Modeling in Medical Imaging

Transformers introduced self-attention as a way to model long-range dependencies (Vaswani et al., 2017). Vision Transformer adapted this idea to image classification by treating image patches as tokens (Dosovitskiy et al., 2021). In medical imaging, UNETR used transformer encoders for 3D segmentation (Hatamizadeh et al., 2022b), while Swin Transformer and Swin UNETR introduced hierarchical shifted-window attention to reduce the cost of global attention (Liu et al., 2021; Hatamizadeh et al., 2022a).

The computational concern with standard self-attention is that the attention matrix grows quadratically with the number of tokens. For an input sequence of length $N$, the attention matrix has size $N \times N$, and the simplified attention operation is:

$$\text{Attention}(Q,K,V) = \left[\text{softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}}\right)\right] V. \tag{4.8}$$

Here, $d_k$ is the key-vector dimension. The square brackets make the scope explicit: softmax acts on the complete scaled score matrix $QK^{\top}/\sqrt{d_k}$, and the resulting attention-weight matrix is then multiplied by $V$. This equation shows why sequence length matters: the multiplication $QK^{\top}$ compares every token with every other token. Window-based methods reduce this cost, but the broader issue remains important for volumetric segmentation because 3D images can produce many spatial tokens.

It would be inaccurate to claim that all transformer models are infeasible or that they are unsuitable for medical imaging. Prior work shows that transformer-based segmentation can be effective. The more careful motivation for this thesis is narrower: for a master's project focused on lightweight deployment and reproducible experiments on a consumer GPU, it is reasonable to investigate alternatives that attempt to combine broader context modeling with lower parameter count.

## 4.5 State Space Models and Mamba-Inspired Design

State Space Models (SSMs) represent sequence dynamics through a hidden state. A continuous-time linear state space model can be written as:

$$h'(t) = Ah(t) + Bx(t), \qquad y(t) = Ch(t). \tag{4.9}$$

Here, $h(t)$ is the hidden state, $x(t)$ is the input signal, $y(t)$ is the output signal, and $A$, $B$, and $C$ are state transition and projection matrices. In the context of deep learning, the appeal of SSMs is that they can model long sequences without explicitly constructing a full $N \times N$ attention matrix.

Mamba extends this line of work through selective state space modeling and hardware-aware sequence processing (Gu and Dao, 2023). Recent biomedical segmentation studies have also explored Mamba-style modules for medical image segmentation. U-Mamba combines convolutional processing with state-space blocks, while SegMamba studies long-range sequential modeling for 3D medical segmentation (Ma et al., 2024; Xing et al., 2024). VM-UNet uses Visual State Space blocks in a U-shaped architecture, Mamba-UNet uses a VMamba-based encoder-decoder and reports experiments on ACDC and Synapse, and LightM-UNet uses residual Vision Mamba layers to pursue a lightweight design (Ruan et al., 2024; Wang et al., 2024; Liao et al., 2024). These works make Mamba/SSM segmentation directly relevant to this thesis, but they are related methods rather than evidence that every Mamba-inspired block will automatically improve cardiac segmentation.

However, this thesis must distinguish the original Mamba implementation, the Visual State Space or SSM blocks in those recent segmentation models, and the implemented project model. Nano-Mamba serializes compressed 3D features and applies only kernel-3 depthwise sequence convolution, token-wise scalar gating, projections, and a residual connection. It has no recurrent state-space transition, selective scan, attention matrix, or direct global all-token interaction. Its fixed raster order also creates boundary adjacencies that are not isotropic 3D neighbours. It is therefore more accurate to describe the contribution as a local Mamba-inspired gated bottleneck exploration, not as a reproduction of Mamba or the cited Visual State Space architectures.

This distinction matters for scientific integrity. If the implementation does not include the full selective scan algorithm, the thesis should not claim that it has implemented the complete Mamba architecture. The contribution is still valid, but it should be framed correctly: the project investigates whether a compact sequence-inspired bottleneck can provide a useful accuracy-efficiency trade-off in cardiac MRI segmentation.

## 4.6 Full-Cine Temporal Segmentation and Cardiac Motion Analysis

Cardiac cine analysis can use time in fundamentally different ways. Qin et al. (2018) jointly learned cardiac MR segmentation and motion estimation using recurrent spatial transformers, thereby producing explicit inter-frame motion information in addition to masks. Yan et al. (2019) used optical-flow learning to add the temporal dimension to cine MRI analysis. Such approaches seek a deformation field or pixel/tissue correspondence between phases; when appropriately validated, this supports motion tracking beyond independent mask measurements.

The present thesis addresses a narrower downstream problem. The trained Nano-Mamba backbone receives one 3D phase at a time and has no temporal input. After all phases are inferred, a fixed circular filter combines class probabilities as $0.25P_{t-1} + 0.50P_t + 0.25P_{t+1}$, including wrap-around at the first and last phases. The resulting masks support LV/RV/MYO volume curves, phase timing, EF, LV-centroid displacement, and a myocardial radial-distance summary. These are segmentation-derived global temporal and motion surrogates. They do not establish point correspondence, a dense deformation field, optical flow, local myocardial displacement, or strain. This distinction defines both the contribution and the validation boundary of the full-cine stage.

### 4.7 Loss Functions and Evaluation Metrics

Medical segmentation is often class-imbalanced because background voxels dominate the image volume, while structures such as myocardium occupy a smaller region. Dice-based objectives are commonly used because they directly optimize overlap between prediction and ground truth (Milletari et al., 2016; Sudre et al., 2017). For a predicted binary mask $P$ and ground-truth mask $G$, the Dice Similarity Coefficient is:

$$\mathrm{DSC}(P,G) = \frac{2|P \cap G|}{|P| + |G|}. \tag{4.10}$$

In multi-class segmentation, Dice can be computed separately for RV, MYO, and LV, and then averaged. This is the reporting strategy used in this thesis.

Cross-entropy is also widely used because it provides voxel-wise classification supervision. If $y_i$ is the true class at voxel $i$ and $p_{i,c}$ is the predicted probability for class $c$, the multi-class cross-entropy can be expressed as:

$$L_{\mathrm{CE}} = -\sum_i \sum_c \mathbf{1}(y_i = c) \log p_{i,c}. \tag{4.11}$$

The final experiment uses MONAI's Dice-Cross Entropy loss, combining region-overlap supervision with voxel-wise classification supervision. This choice is practical rather than novel; it follows established segmentation practice and is supported by the MONAI framework (Cardoso et al., 2022).

Evaluation should also avoid leakage. If the same patient contributes samples to both training and validation, the measured Dice may overestimate generalization. Therefore, the final experimental design of this thesis uses patient-level splitting. This methodological point is as important as the architecture itself, because a strong model comparison is only meaningful when all models are trained and evaluated under the same split.

## 4.8 Summary of the Literature Gap

The literature shows that U-Net-style models remain strong foundations for biomedical segmentation, attention mechanisms can improve feature selection, residual models provide strong convolutional baselines, and transformer/SSM approaches offer explicit mechanisms for longer-range modeling. Recent Mamba segmentation studies include full Visual State Space or state-space components, whereas the present block is a substantially simpler local gated operation. The first gap addressed here is therefore not the absence of segmentation models, but the need to evaluate this compact implementation honestly under a reproducible patient-level protocol.

The motion literature also distinguishes segmentation-derived trajectories from learned motion fields and optical flow. The second gap addressed here is a bounded, auditable bridge from one validated spatial checkpoint to complete-cine global function and motion summaries, together with a direct comparison of frame-wise inference and fixed circular probability fusion. Based on this review, Nano-Mamba U-Net is positioned as a lightweight Mamba-inspired extension of a 3D U-Net-style model embedded in a complete but non-dense temporal analysis. The appropriate questions are whether it offers a useful accuracy–efficiency trade-off on ACDC and what its segmentation outputs can support without claiming learned temporal representation or tissue tracking.

# CHAPTER 5: RESEARCH DESIGN AND METHODOLOGY

This chapter describes the research design used to implement and evaluate the proposed Nano-Mamba U-Net and the subsequent full-cine analysis. The methodology is written to make both stages reproducible: it specifies the dataset boundary, data discovery procedure, patient-level split, preprocessing pipeline, model architecture, loss function, checkpoint selection, evaluation metrics, computational profiling, full-cine inference, temporal probability fusion, and segmentation-derived cardiac measurements. The chapter also distinguishes clearly between methods adopted from prior work and components implemented in this project.

## 5.1 Overall Research Workflow

The study follows a two-stage experimental computing workflow. First, the ACDC cardiac MRI data are discovered and paired with corresponding segmentation labels. Patients are separated into training and validation groups using a deterministic patient-level split. Six spatial segmentation configurations then use the same preprocessing, patient split, epoch budget, optimizer family, and validation procedure, while memory-constrained batch sizes differ by model as reported below. The best checkpoint for each model is selected by validation mean Dice and evaluated descriptively on the same held-out validation patients. Second, the selected Nano-Mamba checkpoint is frozen and applied to every frame of the complete cine sequences for the 20 validation patients. Frame-wise probabilities and fixed temporal fusion are converted to native-grid masks and used to derive cardiac-cycle trajectories and functional measurements. No network is retrained in the second stage.

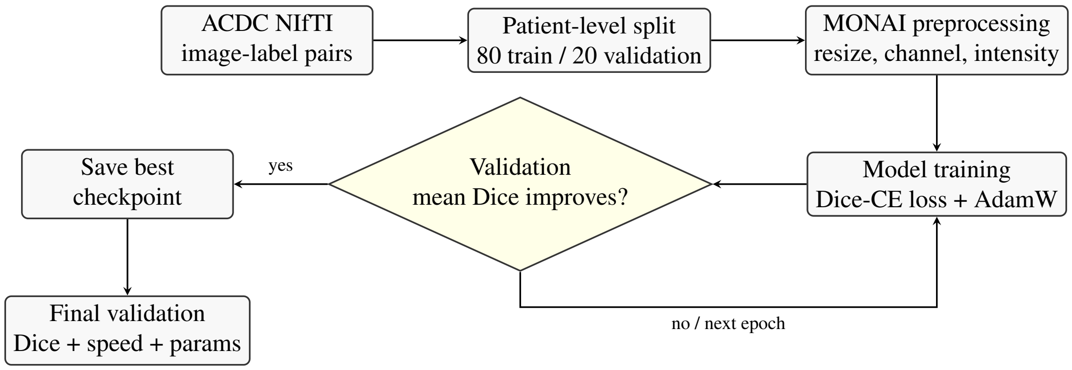


**Figure 5.1: Reproducible experimental workflow. The main methodological safeguard is patient-level separation before training, so that validation cases come from patients not seen during optimization.**

## 5.2 Dataset and Data Discovery

The experiment uses the Automated Cardiac Diagnosis Challenge (ACDC) dataset (Bernard et al., 2018). The task is multi-class segmentation of three anatomical structures: RV, MYO, and LV. The final discovery snapshot records 200 non-empty image-label path pairs from 100 patients. Each patient contributes labelled end-diastolic and end-systolic frames, giving two labelled 3D cases per patient.

A practical issue in this project is that the local dataset layout contains paths ending in `.nii` that may act as containers rather than directly readable NIfTI files. Therefore, the data discovery procedure checks whether each candidate path is a non-empty file or a directory containing a non-empty NIfTI path. Empty placeholders are skipped. It also requires exactly two labelled cases for every patient from `patient001` through `patient100`; duplicate case identifiers or incomplete patient coverage cause the pipeline to stop before splitting. This prevents a partially discovered dataset from silently entering the experiment.

## 5.3 Patient-Level Training and Validation Split

The split was performed at patient level rather than case level. This means all labelled frames from the same patient belong either to training or to validation, never both. This design prevents leakage of patient-specific anatomy into the validation set. With random seed 42 and an 80/20 split, the experiment used 80 training patients and 20 validation patients, corresponding to 160 training cases and 40 validation cases.

Let **P** denote the set of all patient identifiers. The split can be expressed as:

$$\mathbf{P}_{\text{train}} \cap \mathbf{P}_{\text{val}} = \varnothing, \qquad \mathbf{P}_{\text{train}} \cup \mathbf{P}_{\text{val}} = \mathbf{P}. \tag{5.1}$$

This condition is the core methodological safeguard of the experiment. It ensures that the validation Dice measures performance on unseen patients rather than on additional frames from patients used during training.

## 5.4 Preprocessing Pipeline

All models use the same deterministic preprocessing pipeline implemented with MONAI (Cardoso et al., 2022). The exact executed order is:

1. `LoadImaged` loads the paired NIfTI intensity volume and integer label volume.
2. `EnsureChannelFirstd` changes an array shaped $H \times W \times D$ into $1 \times H \times W \times D$.
3. `Resized` maps both arrays to $256 \times 256 \times 16$: trilinear interpolation is used for the continuous MRI intensity and nearest-neighbour interpolation for categorical labels.
4. `ScaleIntensityd` linearly maps the minimum and maximum intensity of the single-channel volume to 0 and 1.
5. `ToTensord` converts both arrays to PyTorch tensors for the data loader.

No random crop, rotation, elastic deformation, intensity augmentation, or other data augmentation is applied. Training and validation use the same deterministic geometric and intensity transforms; only the training data loader shuffles case order. Applying the same preprocessing to all models is necessary because otherwise observed performance differences could be caused by data handling rather than architecture.

The executed transform resizes array dimensions directly; it does not explicitly canonicalize image orientation or resample every case to a common physical voxel spacing before resizing. It also does not crop around the heart. Therefore, the historical six-model training and Dice comparison occur on the common resized array grid and constitute a fixed-array-shape comparison rather than a native-space geometric evaluation. In the later full-cine analysis, predicted class masks are additionally restored to each original spatial array shape with nearest-neighbour interpolation before physical volumes and distances are calculated from NIfTI header voxel sizes. This later inverse resizing enables native-grid measurements but does not retrospectively change the training experiment.

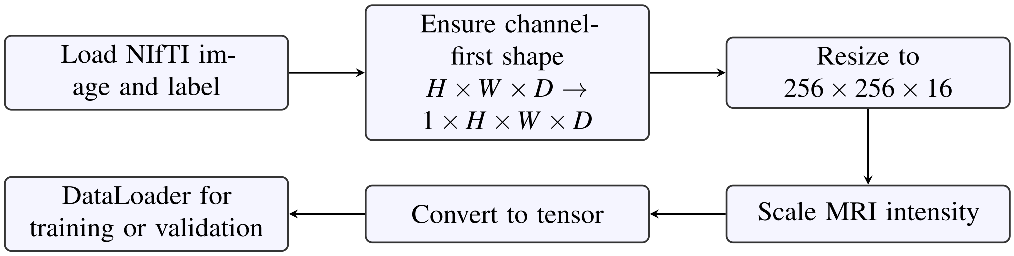


**Figure 5.2: Preprocessing pipeline used for all compared models. Images are interpolated continuously, while labels are resized with nearest-neighbour interpolation to preserve class identities.**

For continuous MRI intensity volumes, trilinear interpolation is used during resizing. If $V(x,y,z)$ denotes the interpolated intensity at a continuous coordinate, the interpolation can be described as a weighted combination of its eight neighbouring voxels:

$$V(x,y,z) \approx \sum_{i=0}^{1}\sum_{j=0}^{1}\sum_{k=0}^{1} w_{ijk} V_{ijk}. \tag{5.2}$$

Here, $V_{ijk}$ are neighbouring voxel intensities and $w_{ijk}$ are interpolation weights determined by the relative distance between the target coordinate and the neighbouring grid points. For segmentation labels, nearest-neighbour interpolation is used instead, because labels are categorical class identifiers rather than continuous intensity values.

MRI intensity values are relative and scanner-dependent rather than standardized physical units (Nyul and Udupa, 1999). Therefore, intensity scaling is applied before network input. For a non-constant single-channel volume, the executed min-max mapping can be written as:

$$S(I_{xyz}) = \frac{I_{xyz} - \min(I)}{\max(I) - \min(I)}. \tag{5.3}$$

This maps the volume minimum to 0 and maximum to 1; MONAI handles the constant-volume edge case. The scaling is applied to each loaded volume, not across the full dataset. No histogram standardization or scanner-specific normalization is performed.

## 5.5 Nano-Mamba U-Net Architecture

The proposed model follows a 3D U-Net-style encoder-decoder structure. The encoder uses convolutional blocks and pooling to reduce spatial resolution while increasing feature channels. The decoder uses transposed convolutions and skip connections to recover spatial resolution and preserve boundary information. The architectural modification studied in this thesis is the insertion of a lightweight Mamba-inspired bottleneck at the compressed latent representation. For Nano-Mamba U-Net, the main tensor path is $B \times 1 \times 256 \times 256 \times 16$ at input, $B \times 128 \times 32 \times 32 \times 2$ after the third pooling operation and the first bottleneck convolution, and $B \times 4 \times 256 \times 256 \times 16$ at the output. The bottleneck therefore serializes $32 \times 32 \times 2 = 2048$ spatial tokens, each with 128 channels. In this thesis, *lightweight* means that Nano-Mamba has the lowest reported trainable-parameter count while retaining competitive batch-one throughput on the recorded hardware. It does not mean that the model leads every speed, memory, or accuracy metric.

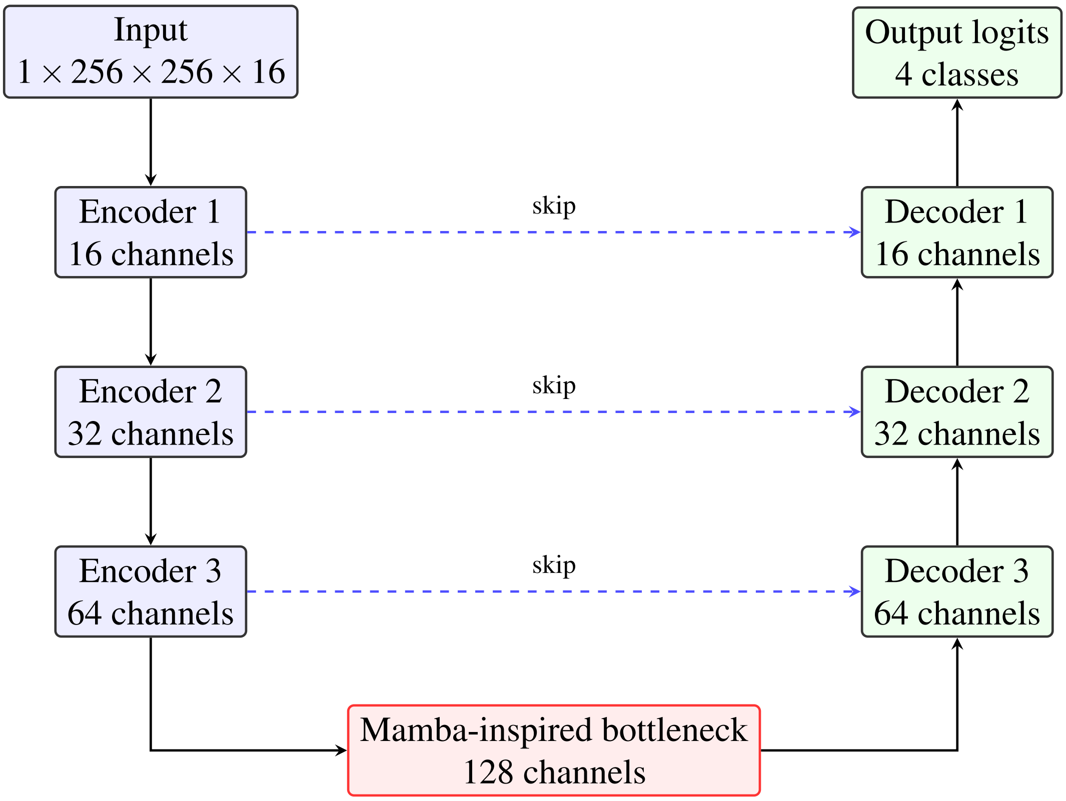


**Figure 5.3: Nano-Mamba U-Net architecture. The model keeps the U-Net encoder-decoder and skip-connection structure, while augmenting the deepest convolutional bottleneck with a compact Mamba-inspired gated sequence module.**

Each convolutional block in Nano-Mamba U-Net and its two custom ablations consists of two 3D convolutional layers followed by batch normalization and ReLU activation. This description does not apply to every MONAI comparison network; their model-specific normalization choices are stated in Section 5.7. A convolutional layer transforms a local neighbourhood into an output feature using learned kernels. For a 3D convolution, this can be written as:

$$Y_{c_o}(p) = \sum_{c_i} \sum_{q \in K} W_{c_o,c_i}(q) X_{c_i}(p+q) + b_{c_o}. \tag{5.4}$$

Here, $X$ is the input feature map, $Y$ is the output feature map, $K$ is the local 3D kernel support, and $W$ and $b$ are trainable parameters. This equation also shows the locality of convolution: each output voxel depends on a neighbourhood around position $p$.

For a five-dimensional `BatchNorm3d` input with shape $N \times C \times D \times H \times W$, the mean and variance for each channel are estimated over the combined $N, D, H, W$ elements, not over the batch axis alone (Ioffe and Szegedy, 2015). For a feature value $x$, batch normalization can be expressed as:

$$\hat{x} = \frac{x - \mu_B}{\sqrt{\sigma_B^2 + \varepsilon}}, \qquad y = \gamma \hat{x} + \beta. \tag{5.5}$$

The parameters $\gamma$ and $\beta$ are learned, allowing the network to recover suitable feature scales after normalization. Consequently, batch size one does not by itself produce an undefined variance when spatial feature elements remain available, although its running statistics come from one patient volume at a time and are a legitimate comparison confound. ReLU activation is then applied as:

$$f(x) = \max(0, x). \tag{5.6}$$

These components are standard neural network operations and are not claimed as novel contributions of this thesis.

## 5.6 Mamba-Inspired Bottleneck Implementation

The bottleneck receives a compressed 3D feature map. Let this tensor be:

$$\mathbf{X} \in \mathbb{R}^{B \times C \times H \times W \times D}. \tag{5.7}$$

The tensor is serialized across the spatial dimensions to obtain a sequence:

$$\mathbf{X}_{\text{seq}} = \text{reshape}(\mathbf{X}) \in \mathbb{R}^{B \times L \times C}, \qquad L = HWD. \tag{5.8}$$

This operation does not create new anatomical or temporal information. It uses a fixed raster order over the spatial grid so that the bottleneck can process compressed volumetric features as a sequence. Here, $D$ is anatomical slice depth, not cardiac time; ED and ES volumes are processed as separate cases.

The implemented bottleneck is Mamba-inspired rather than a full selective scan implementation. It uses linear projection, a symmetric depthwise one-dimensional convolution, a scalar sigmoid gate per token, output projection, and residual addition. The executed graph can be represented as:

$$[\mathbf{U}, \mathbf{V}] = \mathrm{split}\big(\mathrm{Proj}_{in}(\mathbf{X}_{seq})\big)\,, \tag{5.9}$$

$$\mathbf{H} = \mathrm{SiLU}(\mathrm{DWConv1D}(\mathbf{U}))\,, \qquad \mathbf{G} = \mathrm{sigmoid}\Big(\mathrm{Proj}_{\mathrm{gate}}(\mathbf{H})_{:,:,1}\Big)\,, \tag{5.10}$$

$$M_{b\ell c} = H_{b\ell c}\, G_{b\ell 1}\ \mathrm{SiLU}(V_{b\ell c}), \qquad \mathbf{Y} = \mathrm{Proj}_{\mathrm{out}}(\mathbf{M}), \qquad \mathbf{Z} = \mathbf{Y} + \mathbf{X}_{\mathrm{seq}}. \tag{5.11}$$

Here, $\mathrm{Proj}_{\mathrm{in}}$ and $\mathrm{Proj}_{\mathrm{out}}$ are learned linear projections, DWConv1D is the depthwise sequence convolution, and sigmoid is the gate activation. The component equation for $M_{b\ell c}$ states the element-wise multiplication explicitly: batch index $b$, token index $\ell$, and channel index $c$ are not contracted. In the executed implementation, $\mathrm{Proj}_{\mathrm{gate}}$ produces $2d_{state} + 1$ channels but only its first channel enters $\mathbf{G}$. The remaining legacy projection channels are retained for checkpoint compatibility but have no path to the output. The output sequence is reshaped back into 3D feature form before being passed to the decoder.

The executed block performs kernel-3 depthwise mixing along the fixed raster-serialized latent sequence and applies a token-wise scalar gate. It contains no recurrent state-space propagation, selective scan, self-attention, or direct global all-token interaction. Moreover, raster serialization introduces order-dependent adjacencies where rows and slices meet; a pair of consecutive tokens is not always a pair of isotropic 3D neighbours. The operation is therefore best understood as compact local gated sequence mixing at the bottleneck, not as global 3D context modeling.

This design choice must be interpreted carefully. The original Mamba paper proposes selective state space sequence modeling (Gu and Dao, 2023). The implementation in this project uses a lightweight PyTorch block inspired by this direction, but it does not claim to reproduce the full hardware-aware selective scan. The research contribution is therefore the integration and evaluation of a compact sequence-inspired bottleneck inside a cardiac MRI segmentation model.

## 5.7 Comparison Models and Ablation Design

The experiment compares Nano-Mamba U-Net with several reference architectures. The 3D U-Net baseline represents the standard encoder-decoder approach. Attention U-Net represents attention-gated skip connections. SegResNet16 represents a residual convolutional segmentation model based on the SegResNet family (Myronenko, 2019). Two custom controls study the bottleneck. The *convolutional-bottleneck control (No-Mamba)* contains zero Mamba or Mamba-inspired operations: it executes `DoubleConv(64,128)` followed by `DoubleConv(128,128)`. The *64-channel Mamba-bottleneck ablation (Half-Mamba)* executes `DoubleConv(64,64)`, the same type of Mamba-inspired block at 64 channels, and then `DoubleConv(64,128)`. "Half" therefore refers only to the sequence-block channel width relative to Nano's 128-channel block, not to half of the complete network.

The executed normalization and batch-size combinations are model specific:

**Table 5.1: Executed normalization and training batch size by model family.**

| Configuration | Executed normalization | Batch size |
|---|---|---|
| MONAI 3D U-Net | Instance normalization | 2 |
| Nano, convolutional control, and 64-channel ablation | `BatchNorm3d` | 2 |
| MONAI Attention U-Net | `BatchNorm3d` | 1 |
| MONAI SegResNet16 | Group normalization (8 groups) | 1 |

Attention U-Net therefore did execute batch normalization at batch size one. This did not crash: its recovered training log contains all 150 epochs and its best checkpoint is epoch 62. Its three-dimensional batch-normalization layers still aggregate over spatial elements for each channel, but the single-volume statistics may be noisier and cannot be separated from architecture in the reported comparison. SegResNet16 did not have this specific conflict because its default executed normalization was group normalization, which does not use batch statistics.

The ablations are not parameter matched: the convolutional-bottleneck control contains 2.29M reported parameters, the 64-channel Mamba-bottleneck ablation contains 1.64M, and Nano-Mamba U-Net contains 1.46M. The control also has a structurally different two-block convolutional bottleneck. These comparisons describe executed architecture-level accuracy–efficiency trade-offs, but they cannot isolate a causal benefit or harm from sequence gating alone. This distinction is important because the zero-Mamba convolutional control slightly outperformed Nano-Mamba U-Net in mean Dice, while Nano-Mamba retained the lowest reported parameter count.

## 5.8 Training Objective and Optimization

The models are trained using a Dice-Cross Entropy objective. Dice loss addresses region-level overlap, while cross entropy provides voxel-wise class supervision. For class $c$, the Dice score is:

$$\mathrm{DSC}_c = \frac{2\sum_i p_{i,c} g_{i,c}}{\sum_i p_{i,c} + \sum_i g_{i,c} + \varepsilon}. \tag{5.12}$$

The corresponding Dice loss can be written as:

$$L_{\mathrm{Dice}} = 1 - \frac{1}{C}\sum_{c=1}^{C} \mathrm{DSC}_c\,. \tag{5.13}$$

The multi-class cross-entropy term is:

$$L_{\text{CE}} = -\sum_i \sum_c g_{i,c} \log(p_{i,c}). \tag{5.14}$$

The total loss used for optimization is:

$$L_{\text{total}} = L_{\text{Dice}} + L_{\text{CE}}. \tag{5.15}$$

Here, $p_{i,c}$ denotes the predicted probability for voxel $i$ and class $c$, while $g_{i,c}$ denotes the one-hot ground-truth label. This loss is a standard practical choice for medical segmentation and is implemented using MONAI's Dice-Cross Entropy loss. The executed constructor sets `to_onehot_y=True` and `softmax=True` but does not override `include_background`; under the documented MONAI default, the four-channel training Dice term includes background. By contrast, the reported evaluation Dice is computed only for the three foreground structures. The optimization loss and the reported metric therefore use related but not identical class averaging.

Optimization is performed with AdamW (Loshchilov and Hutter, 2019). The learning rate is $1 \times 10^{-4}$ and weight decay is $1 \times 10^{-5}$. All models are trained for 150 epochs without a learning-rate scheduler or early stopping; validation is run after every epoch. UNet3D, Nano-Mamba, the convolutional-bottleneck control, and the 64-channel Mamba-bottleneck ablation use batch size 2; Attention U-Net and SegResNet16 use batch size 1 because of memory constraints. This is not a strictly controlled batch-size comparison. Different batch sizes imply different numbers of optimizer updates per epoch, and normalization choices also differ across model families. The table should therefore be interpreted as a comparison of executed model configurations, not a perfectly controlled isolation of architecture alone.

## 5.9 Checkpoint Selection and Evaluation Metrics

A key methodological correction in the final experiment is validation-based checkpoint selection. Earlier exploratory scripts saved best models according to training loss, which is not a reliable indicator of generalization. In the final experiment, the checkpoint is saved only when validation mean Dice improves. If $\overline{D}_{val}^{(e)}$ denotes the validation mean Dice at epoch $e$, the selected checkpoint is:

$$e^* = \arg\max_e \overline{D}_{val}^{(e)}. \tag{5.16}$$

The final reported metrics are computed from this selected checkpoint on the held-out validation cases. Because the same validation set is used for repeated checkpoint selection and descriptive reporting, it must not be interpreted as an independent final test set.

The primary reported segmentation metric is mean Dice across RV, MYO, and LV:

$$\overline{D} = \frac{D_{RV} + D_{MYO} + D_{LV}}{3}. \tag{5.17}$$

At inference, the network returns four logit channels on the resized grid. The class with the largest logit is selected voxel-wise using `argmax`; there is no thresholding, connected-component filtering, hole filling, or other mask post-processing. The reference label is resized with nearest-neighbour interpolation and cast to integer class identifiers. Dice is first computed for each foreground class in each case, then averaged across the 40 validation cases; the three class means are averaged to obtain the reported mean Dice.

The implementation assigns a class Dice of 1 when both the predicted and reference masks for that class are empty in a case; otherwise it applies the unsmoothed foreground Dice expression. This convention is stated because it can affect case-level aggregation. The audited evidence contains 40 per-case rows for each of the six models (720 foreground class scores in total). None of those class scores is exactly 1, so the empty-empty branch did not contribute to the reported aggregate values. The convention is nevertheless retained here because it remains part of the executed metric definition.

For post-hoc uncertainty description, the two ED/ES case values are first averaged within each of the 20 validation patients. A patient-level percentile bootstrap then resamples 20 patients with replacement for 10,000 replicates using audit seed 20260820. This produces 95% intervals for each model's patient-macro mean Dice and for paired Nano-Mamba differences. Resampling at patient level preserves the dependence between the two cases from one patient and between model predictions on the same patient. These intervals were added after recovery of the original per-case CSVs; they are descriptive, not pre-registered hypothesis tests. Because the same 20 patients were used repeatedly for checkpoint selection, the intervals do not convert the validation results into independent test-set estimates.

The study also reports parameter count and inference speed. The reported parameter count sums every tensor returned by `model.parameters()`, including the unused legacy outputs of the gate projection described above. For Nano-Mamba U-Net this sum is exactly 1,456,325 trainable parameters, or 1.456325 million. The checkpoint state dictionary contains 1,457,747 scalar entries because it additionally stores 1,422 non-parameter batch-normalization buffers (running means, running variances, and batch counters); this is not a parameter-count discrepancy. Inference speed uses a random batch-one tensor of shape $1 \times 1 \times 256 \times 256 \times 16$, five warm-up passes, and 30 timed passes with CUDA synchronization. These computational metrics are engineering measurements from one implementation and environment; parameter count is not treated as a direct proxy for speed.

### 5.10 Full-Cine Spatio-Temporal Analysis

#### 5.10.1 Cohort, cine discovery, and identity checks

The full-cine stage uses the same 20 validation patients as the historical segmentation experiment. Their complete ACDC cine volumes contain 550 cardiac frames in total. For each patient, `Info.cfg` provides the number of frames, the annotated ED frame, the annotated ES frame, and the diagnostic group. Cine discovery is based on the NIfTI header shape $(X,Y,Z,T)$ and requires $T$ to match `NbFrame`; it does not depend on a fragile file name or file-size assumption.

Before inference, the standalone ED and ES images used by the historical experiment are compared with the corresponding complete-cine frames after the executed intensity normalization. The maximum normalized mean absolute error across the final cohort is zero. Six complete cine files restored from a pinned public ACDC mirror contain raw affine metadata that differs from the standalone endpoint container, although spatial shape, header voxel sizes, and endpoint pixel arrays match, and each endpoint image affine matches its manual label. The pipeline records this discrepancy rather than silently overwriting it. Volumes and distances use the NIfTI header voxel sizes, which agree between the cine and endpoint data.

#### 5.10.2 Frame inference and temporal fusion

For each cardiac phase $t$, the 3D frame $I_t \in \mathbb{R}^{X\times Y\times Z}$ follows the checkpoint-compatible preprocessing path: channel-first conversion, trilinear resizing to $256\times256\times16$, per-frame min–max intensity scaling, and batch-one inference. Softmax converts the four output logits into class probabilities $P_t$. Frame-wise masks are obtained from the original float32 logits. To limit host-memory use, the executed full-cine run stored the probability maps as float16 before converting them back to float32 for the following fusion arithmetic. The pre-specified temporal alternative uses fixed circular adjacent-frame probability fusion:

$$\widetilde{P}_t = 0.25P_{t-1} + 0.50P_t + 0.25P_{t+1}, \tag{5.18}$$

where phase indices wrap at the end of the cardiac cycle. The weight of the current frame is twice that of either neighbour, and the weights sum to one. This operation has no learned parameter and is applied only after the spatial network has generated all frame probabilities. For either method, voxel labels are obtained by channel-wise `argmax`; the categorical mask is then resized back to $(X,Y,Z)$ using nearest-neighbour interpolation. No thresholding, connected-component filtering, hole filling, registration, or deformation estimation is performed.

### 5.10.3 Physical volumes, phase timing, and function

Let $(s_x, s_y, s_z)$ be the spatial voxel sizes in millimetres from the cine header. One voxel represents

$$v_{voxel} = \frac{s_x s_y s_z}{1000}\ \text{mL}. \tag{5.19}$$

If $N_{k,t}$ is the number of voxels assigned to class $k$ at phase $t$, its physical volume is $V_{k,t} = N_{k,t} v_{voxel}$. The complete LV sequence $V_{LV,t}$ forms a segmentation-derived volume trajectory. To reduce one-frame phase jitter only for phase detection, a circular three-frame mean is used:

$$\widehat{V}_{LV,t} = \frac{V_{LV,t-1} + V_{LV,t} + V_{LV,t+1}}{3}. \tag{5.20}$$

Predicted ED is the maximum of $\widehat{V}_{LV,t}$ and predicted ES is its minimum. Phase error is the shortest circular distance in frames to the `Info.cfg` reference. End-diastolic volume, end-systolic volume, stroke volume, and ejection fraction are then

$$EDV = \widehat{V}_{LV,t_{ED}}, \quad ESV = \widehat{V}_{LV,t_{ES}}, \quad SV = EDV - ESV, \tag{5.21}$$

$$EF = 100\frac{EDV - ESV}{EDV}. \tag{5.22}$$

For endpoint validation, a reference EF is also calculated directly from the manual ED and ES LV masks, and a prediction EF is calculated from the predicted masks at those same annotated phases. This isolates segmentation-derived functional error from phase-selection error.

### 5.10.4 Global motion surrogates and temporal consistency

For each frame, the LV-mask centroid is converted into millimetre coordinates using a metric transform constructed from the header voxel sizes. The peak global LV-centroid displacement is

$$d_{peak} = \max_t \left\| \mathbf{c}_{LV,t} - \mathbf{c}_{LV,ED} \right\|_2. \tag{5.23}$$

The mean radial distance of myocardial voxels from the LV centroid is also computed at every frame. The reported ED-minus-ES radial-distance change is a global contraction surrogate. Neither quantity is a dense displacement field, regional strain estimate, or material-point tracker because successive masks do not establish voxel correspondence.

Temporal curve smoothness is quantified by the mean absolute circular second difference normalized by the smoothed LV-curve amplitude:

$$S = \frac{1}{T\left[\max(\widehat{V}) - \min(\widehat{V})\right]} \sum_{t=1}^{T} \left| \widehat{V}_{t+1} - 2\widehat{V}_t + \widehat{V}_{t-1} \right|. \tag{5.24}$$

A smaller value means less frame-to-frame curvature in the global volume trajectory; it does not by itself prove anatomical correctness.

### 5.10.5 Validation and uncertainty

Manual segmentation masks are available only at ED and ES. Both resized-grid endpoint Dice, which connects to the historical experiment, and native-grid endpoint Dice after nearest-neighbour mask restoration are reported. EF error is reported in percentage points, and ED/ES phase detection is reported as exact-frame and within-one-frame accuracy. Patient-level means and paired temporal-fusion-minus-frame-wise differences use 10,000 bootstrap replicates with seed 20260821. The analysis also reports diagnostic-group summaries, but their group sizes range from one to seven patients and are therefore descriptive only. Every frame row, patient row, figure, source hash, checkpoint hash, and aggregate value is preserved in the result bundle and independently recomputed by a second audit script.

## 5.11 Hardware and Reproducibility

The experiment used PyTorch (Paszke et al., 2019) and MONAI on a single NVIDIA RTX 4060 GPU with 8GB VRAM. Seed 42 is applied to Python's random generator, NumPy, PyTorch on the CPU, and PyTorch on every CUDA device. The executed entry point also enables the CuDNN deterministic flag, disables the CuDNN benchmark flag, and gives both data loaders zero worker processes. Thus, none of the seed or CuDNN controls stated in this method is missing from the entry point. The historical run did not enable the PyTorch mode that rejects nondeterministic operations, so the correct claim is controlled seeding and deterministic CuDNN configuration rather than guaranteed bitwise reproducibility. All six result rows are accompanied by 40-case metric tables and 150-epoch training logs; their class averages and selected best epochs reproduce the aggregate result table. The full-cine run used Python 3.10.20, PyTorch 2.7.1 with CUDA

11.8, MONAI 1.5.2, the epoch-121 Nano-Mamba checkpoint, and batch size one. The patient split, preprocessing code, model definitions, evaluation code, full-cine result rows, figures, and hashes are retained in the project repository. Deterministic settings and sealed artifacts improve repeatability within a fixed software and hardware stack, but they do not guarantee bitwise identity across library versions or devices.

### 5.12 Methodological Summary

The methodology supports a cautious six-model comparison and a separate full-cine analysis under one patient-level split. It does not provide an independent test set, matched batch sizes, matched-capacity ablations, multi-seed training, augmentation, intermediate-frame manual labels, or external clinical validation. Historical six-model Dice remains a resized-grid measure; the full-cine stage additionally reports native-grid endpoint Dice and physical measurements. The backbone is not a full implementation of the original Mamba selective scan, and fixed probability fusion is not learned temporal modeling. The study therefore addresses both the compact accuracy–efficiency trade-off and a bounded segmentation-derived cardiac-cycle analysis, not universal architectural superiority or dense motion tracking.

# CHAPTER 6: RESULTS

This chapter reports the empirical results obtained from the revised patient-level validation experiment. Unlike the earlier exploratory scripts, the final experiment used a deterministic 80/20 patient-level split of the ACDC training cohort. The split contained 80 training patients and 20 held-out validation patients, corresponding to 160 training cases and 40 validation cases. The best checkpoint for each model was selected using validation mean Dice, and the final quantitative results were reported only on the held-out validation cases.

The objective of this chapter is descriptive rather than argumentative. It reports what the experiment shows before Chapter 7 interprets the implications. This separation is important because the final results are more nuanced than a simple claim of superiority. Nano-Mamba U-Net is not the highest-Dice model in the final experiment. Its value lies in the combination of competitive segmentation performance and the smallest parameter count among the evaluated models.

## 6.1 Experimental Setting

All models were trained and evaluated using the same deterministic preprocessing and validation protocol. The input volumes were resized to $256 \times 256 \times 16$, intensity-scaled to $[0, 1]$ per volume, and converted to PyTorch tensors using MONAI transforms (Cardoso et al., 2022). Images used trilinear interpolation, labels used nearest-neighbour interpolation, and no data augmentation or mask post-processing was applied. The evaluated anatomical classes were the Right Ventricle (RV), Myocardium (MYO), and Left Ventricle (LV). The primary segmentation metric was the Dice Similarity Coefficient (DSC), computed on the resized grid.

The comparison included a standard 3D U-Net baseline, Attention U-Net, SegResNet16, the proposed Nano-Mamba U-Net, a zero-Mamba convolutional-bottleneck control, and a 64-channel Mamba-bottleneck ablation. SegResNet16 refers to the configuration with `init_filters=16`, which was the version successfully trained under the final patient-level validation protocol. Therefore, the parameter count reported in this chapter is 4.70M for SegResNet16, not the larger 18.80M configuration explored in earlier standalone benchmarking scripts.

For all models, the validation mean Dice was computed as the average of the three foreground classes:

$$\overline{D}_{val} = \frac{D_{RV} + D_{MYO} + D_{LV}}{3}. \tag{6.1}$$

This formulation avoids allowing one strong class, such as LV, to dominate the interpretation. Reporting RV, MYO, and LV separately is necessary because a clinically useful segmentation model should not only perform well on the easiest structure.

## 6.2 Quantitative Validation Results

Table 6.1 presents the final held-out validation performance and computational statistics. The results show that SegResNet16 achieved the highest validation mean DSC of 86.70%. The proposed Nano-Mamba U-Net achieved a mean DSC of 84.78% with the smallest parameter count among the evaluated models, at 1.46M parameters.

**Table 6.1: Held-out patient-level validation results on the ACDC training cohort. Intervals are post-hoc patient-level percentile-bootstrap 95% intervals (10,000 replicates); the same validation set was used for checkpoint selection.**

| Architecture | RV (%) | MYO (%) | LV (%) | Mean DSC (%) | 95% CI (%) | Params (M) | FPS |
|---|---|---|---|---|---|---|---|
| 3D U-Net Baseline | 78.35 | 74.57 | 89.59 | 80.83 | [77.78, 83.46] | 4.81 | 88.17 |
| Attention U-Net | 69.43 | 72.38 | 82.53 | 74.78 | [65.37, 82.44] | 5.91 | 22.41 |
| SegResNet16 | **84.30** | **82.76** | **93.03** | **86.70** | [84.26, 88.55] | 4.70 | 25.50 |
| Conv. control (No-Mamba) | 83.26 | 81.30 | 92.37 | 85.64 | [83.08, 87.64] | 2.29 | 28.80 |
| 64-ch Mamba ablation (Half) | 82.41 | 80.46 | 91.99 | 84.95 | [81.85, 87.21] | 1.64 | 29.04 |
| Nano-Mamba U-Net | 82.11 | 80.35 | 91.88 | 84.78 | [82.05, 86.90] | **1.46** | 29.09 |

Compared with the 3D U-Net baseline, Nano-Mamba U-Net improved the mean DSC by 3.95 percentage points while reducing the parameter count from 4.81M to 1.46M.

Using the unrounded parameter counts from the result CSV, the reduction relative to 3D U-Net is:

$$\frac{4.808537 - 1.456325}{4.808537} \times 100\% \approx 69.7\%. \tag{6.2}$$

Compared with SegResNet16, Nano-Mamba U-Net used approximately 69.0% fewer parameters, but its mean DSC was 1.92 percentage points lower. These results indicate that Nano-Mamba U-Net is not the highest-accuracy model in the final experiment; instead, its main advantage lies in parameter efficiency with competitive validation performance.

Table 6.2 reports paired patient-level differences. Positive values favor Nano-Mamba U-Net. Pairing is important because every model was evaluated on the same 20 patients. The intervals are descriptive resampling summaries; no $p$-values, multiplicity correction, or independent test cohort is claimed.

**Table 6.2: Post-hoc paired patient-level Nano-Mamba mean-DSC differences.**

| Reference model | Nano minus reference (pp) | 95% bootstrap CI (pp) |
|---|---:|:---:|
| 3D U-Net Baseline | +3.95 | [+2.83, +5.01] |
| Attention U-Net | +10.00 | [+3.51, +18.29] |
| SegResNet16 | -1.92 | [-3.01, -0.88] |
| Conv. control (No-Mamba) | -0.86 | [-1.48, -0.23] |
| 64-ch Mamba ablation (Half) | -0.17 | [-0.91, +0.55] |

### 6.3 Accuracy-Efficiency Visualization

Since the research question concerns both segmentation accuracy and model efficiency, the results should not be interpreted from Table 6.1 alone. Figure 6.1 visualizes the relationship between validation mean Dice and trainable parameter count. A model positioned higher on the vertical axis has stronger Dice performance, while a model positioned further left has fewer parameters.

This visualization makes the central interpretation clearer. SegResNet16 is the accuracy leader and should be acknowledged as such. Nano-Mamba U-Net is not located at the top of the graph, but it appears on the left side because it has the smallest parameter count.

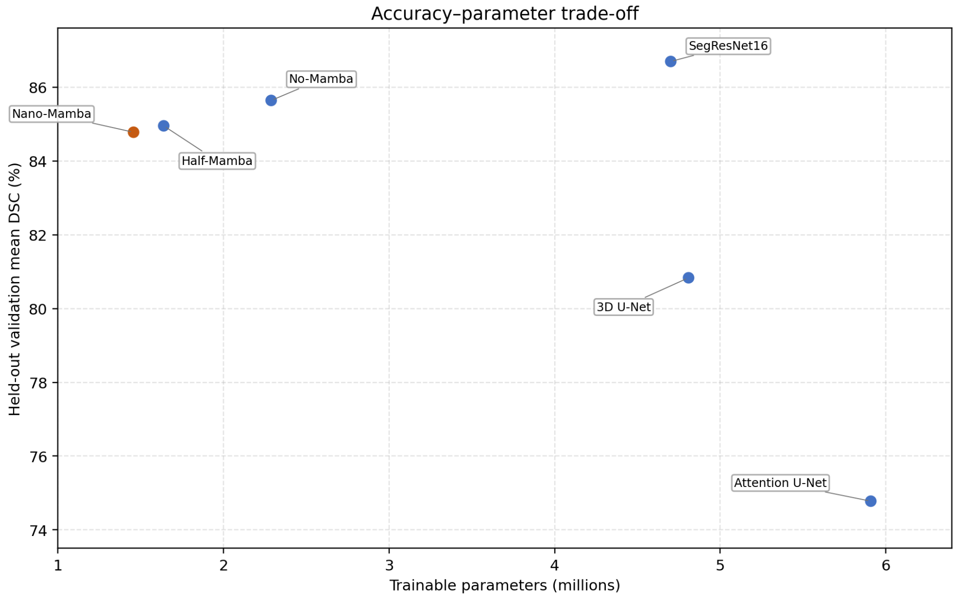


**Figure 6.1: Accuracy-parameter trade-off generated directly from the audited summary CSV. SegResNet16 obtains the highest held-out validation Dice score, while Nano-Mamba U-Net occupies the lowest-parameter region.**

The result therefore supports an efficiency-oriented contribution rather than an absolute accuracy claim.

Figure 6.2 visualizes the relationship between validation mean Dice and inference speed. The speed result should be interpreted carefully because FPS depends on implementation details, hardware, batch size, input resolution, and measurement protocol. Therefore, FPS is reported as a practical engineering indicator rather than as a universal property of the architecture.

## 6.4 Model-by-Model Interpretation

The 3D U-Net baseline achieved 80.83% mean DSC. This result is important because it provides a conventional volumetric segmentation reference point. Nano-Mamba U-Net improved over this baseline on all three structures, suggesting that the proposed compact architecture is not merely smaller but also more accurate than this particular 3D U-Net configuration under the final split.

Attention U-Net achieved the lowest mean DSC among the evaluated models, at 74.78%.

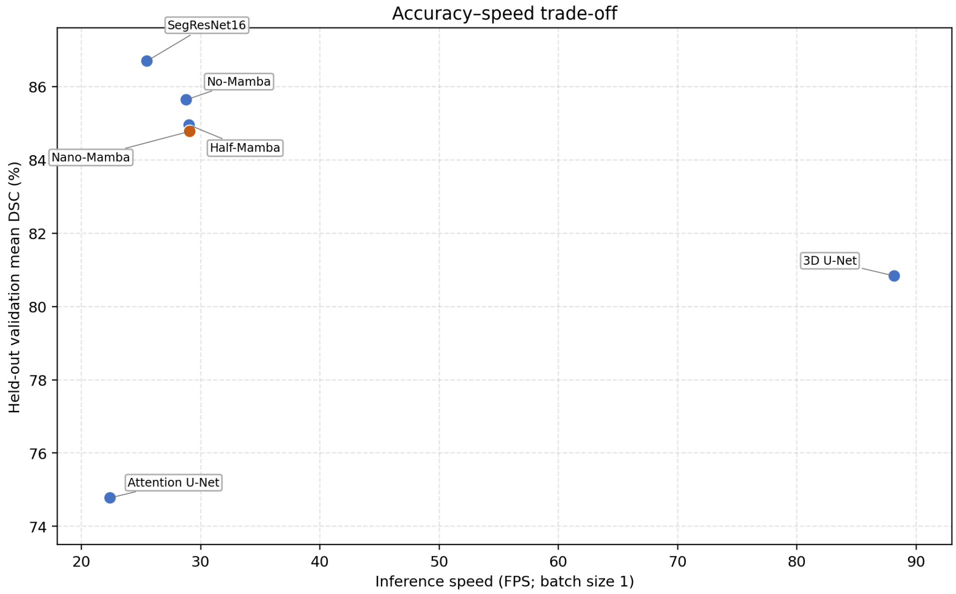


**Figure 6.2: Accuracy-speed trade-off generated directly from the audited summary CSV. The figure provides an engineering view of held-out validation Dice relative to batch-one inference speed in the recorded experiment.**

This does not mean attention gates are generally ineffective. It means that, under the implemented settings and validation split, this Attention U-Net configuration did not train or generalize as strongly as the other models. This result also demonstrates why it is unsafe to assume that a more complex architectural concept necessarily improves performance.

SegResNet16 achieved the highest mean DSC, with the best RV, MYO, and LV scores. This suggests that residual convolutional modeling remains very strong for this dataset. Since SegResNet16 outperformed Nano-Mamba U-Net by 1.92 percentage points, the thesis must not describe Nano-Mamba U-Net as the best model by Dice. Instead, SegResNet16 should be acknowledged as the accuracy leader in the final experiment.

The convolutional-bottleneck control (No-Mamba) achieved 85.64% mean DSC, which is 0.86 percentage points higher than Nano-Mamba U-Net. Despite its historical name, this control contains no Mamba or Mamba-inspired operation. The paired post-hoc interval for Nano minus the control is $[-1.48, -0.23]$ percentage points. This is important negative evidence against claiming that adding the present gated block increases Dice. The control is not parameter matched and uses 2.29M parameters compared with Nano-Mamba's 1.46M, so the comparison does not isolate sequence gating as the cause of the difference. It shows only that the executed convolutional-control architecture obtained higher Dice on this selected split while Nano-Mamba retained 36.3% fewer parameters and essentially the same recorded FPS.

The 64-channel Mamba-bottleneck ablation (Half-Mamba) achieved 84.95% mean DSC, 0.174 percentage points above Nano-Mamba U-Net. "Half" describes the Mamba-inspired block's channel width, not half of the entire model. Despite the wider sequence block in Nano-Mamba, its overall architecture used 1.456M reported trainable parameters versus 1.638M for the 64-channel ablation. Nano-Mamba therefore used approximately 11.1% fewer parameters while retaining 99.8% of the ablation's mean DSC. The paired interval for Nano minus this ablation is $[-0.91, +0.55]$ percentage points and crosses zero, so the available case-level evidence does not establish either a reliable accuracy penalty for Nano-Mamba or a reliable accuracy advantage for the 64-channel ablation. This is a compactness result rather than an equivalence claim; a matched-capacity, multi-seed experiment would be required to study bottleneck size more reliably.

## 6.5 Class-Specific Performance

Across most models, LV segmentation achieved the highest Dice score, while MYO segmentation was generally more difficult. This pattern is consistent with the anatomical properties of cardiac MRI: the LV cavity is usually more clearly defined, whereas the myocardium is a thinner ring-like structure affected by papillary muscles, partial volume effects, and boundary ambiguity.

For Nano-Mamba U-Net, the LV Dice was 91.88%, the RV Dice was 82.11%, and the

MYO Dice was 80.35%. The gap between LV and MYO is:

$$91.88\% - 80.35\% = 11.53\%. \tag{6.3}$$

This difference is clinically plausible because the myocardium is more difficult to delineate consistently. A small absolute boundary shift can affect a thin myocardial ring more strongly than a larger ventricular cavity. Therefore, the lower MYO score should be interpreted as a known structural challenge rather than simply as a random model failure.

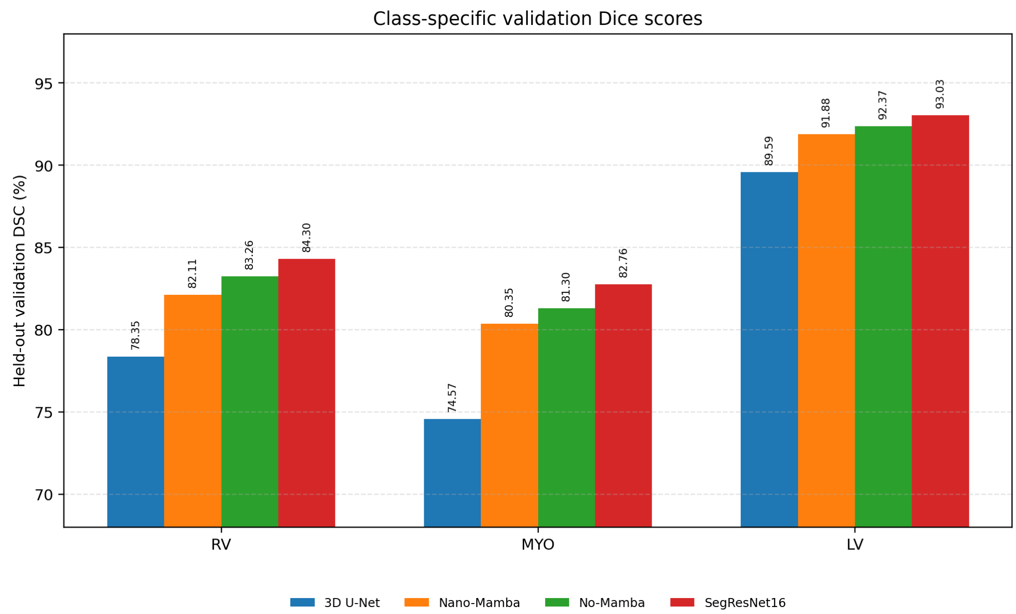


**Figure 6.3: Class-specific held-out validation Dice for four representative models selected to show the basic 3D U-Net baseline, Nano-Mamba U-Net, its convolutional-bottleneck control, and the highest-Dice SegResNet16 model. The complete six-model comparison remains in Table 6.1.**

## 6.6 Checkpoint Selection Evidence

The experiment logs contain 150 contiguous epoch rows for every model. For each log, the maximum recorded validation mean DSC occurs at exactly the best epoch in the aggregate table: epochs 105, 121, 134, 149, 62, and 112 for 3D U-Net, Nano-Mamba, the convolutional control (No-Mamba), the 64-channel ablation (Half-Mamba), Attention

U-Net, and SegResNet16, respectively. Figure 6.4 therefore shows the recorded raw validation traces rather than hand-entered points.

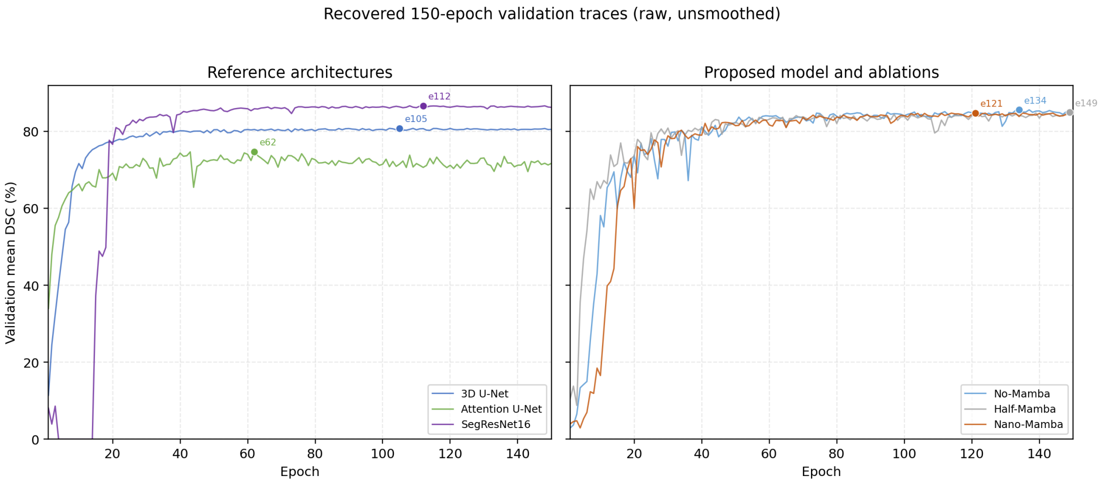


**Figure 6.4: Raw, unsmoothed 150-epoch validation mean-DSC traces. Markers identify the best validation checkpoint selected for each model. The figure is generated directly from the six experiment-log CSVs.**

The learning curves also show why validation-based checkpoint selection was necessary. By epoch 150, the validation score was 0.29 percentage points below the selected maximum for both Nano-Mamba and 3D U-Net, 0.43 points lower for SegResNet16, 0.56–0.62 points lower for the two ablations, and 3.05 points lower for Attention U-Net. Training loss nevertheless continued to decrease for all six models. The larger Attention U-Net gap is the clearest sign of late-epoch overfitting or instability in the recorded runs; it does not imply that attention mechanisms are generally inferior.

The per-case tables identify a shared difficult case. `patient049_frame11` is the lowest-mean-Dice case for Nano-Mamba, 3D U-Net, SegResNet16, the convolutional control, and the 64-channel ablation. Nano-Mamba obtains 57.86% mean DSC on this case, driven mainly by low RV Dice of 26.04%. Attention U-Net's worst case is `patient042_frame16`, at 16.38% mean DSC. Because no pathology label or visual overlay is used in this analysis, these numerical failures should not be assigned an anatomical cause without inspecting the corresponding validation images.

### 6.7 Full-Cine Spatio-Temporal Results

#### 6.7.1 Coverage and endpoint consistency

The separate full-cine experiment successfully processed all 20 validation patients and all 550 recorded cardiac phases. This produced 1,100 per-frame rows because each frame was evaluated by both frame-wise inference and fixed temporal probability fusion, together with 40 patient-level rows. Before inference, every complete-cine ED/ES frame reproduced its corresponding standalone endpoint image with maximum normalized mean absolute error of zero. Frame-wise endpoint Dice also reproduced the historical Nano-Mamba case table with a maximum absolute difference of 0.000169. These checks establish that the cine analysis used the intended validation patients, endpoint phases, checkpoint, image contents, and class convention.

#### 6.7.2 Endpoint segmentation, EF, and curve smoothness

Table 6.3 summarizes the main full-cine comparison. The resized-grid endpoint Dice is reported to connect the analysis to the historical experiment. Native-grid Dice is lower because categorical masks are resized back to the original spatial shape before comparison; this exposes the effect of the non-native training grid rather than hiding it.

**Table 6.3: Full-cine results for independent frame-wise inference and fixed temporal probability fusion. Paired intervals use 10,000 patient-level bootstrap replicates.**

| Metric | Frame-wise | Temporal fusion | Fusion minus frame-wise | 95% paired CI |
|---|---|---|---|---|
| Endpoint Dice, resized grid (%) | 84.779 | 84.794 | +0.014 pp | [-0.063, +0.082] pp |
| Endpoint Dice, native grid (%) | 77.919 | 77.923 | +0.005 pp | not estimated |
| Annotated-phase EF MAE (pp) | 3.514 | 3.329 | -0.185 pp | [-0.468, +0.091] pp |
| Normalized LV-curve second difference | 0.03252 | 0.03139 | -0.001129 | [-0.001950, -0.000511] |

Temporal fusion increased resized-grid endpoint Dice for 13 of 20 patients and decreased it for seven. Its paired Dice interval includes zero, so the observed increase of 0.014 percentage points is not a resolved accuracy improvement. EF absolute error decreased for 11 patients and increased for nine; this paired interval also includes zero. In contrast, the normalized LV-curve second difference decreased for 19 of 20 patients, and its paired interval is entirely below zero. The supported conclusion is therefore narrow but positive: fixed adjacent-frame fusion made the segmentation-derived LV trajectories consistently smoother on this cohort without establishing a clear endpoint-Dice or EF advantage.

Figure 6.5 shows both population-normalized LV trajectories and their paired pointwise difference. Time is normalized only for the group visualization; patient-level phases and metrics use the original frame indices. The two mean curves nearly overlap, which is consistent with the very small aggregate changes. The lower panel exposes the small fusion-minus-frame-wise difference without implying a resolved endpoint-Dice or EF benefit.

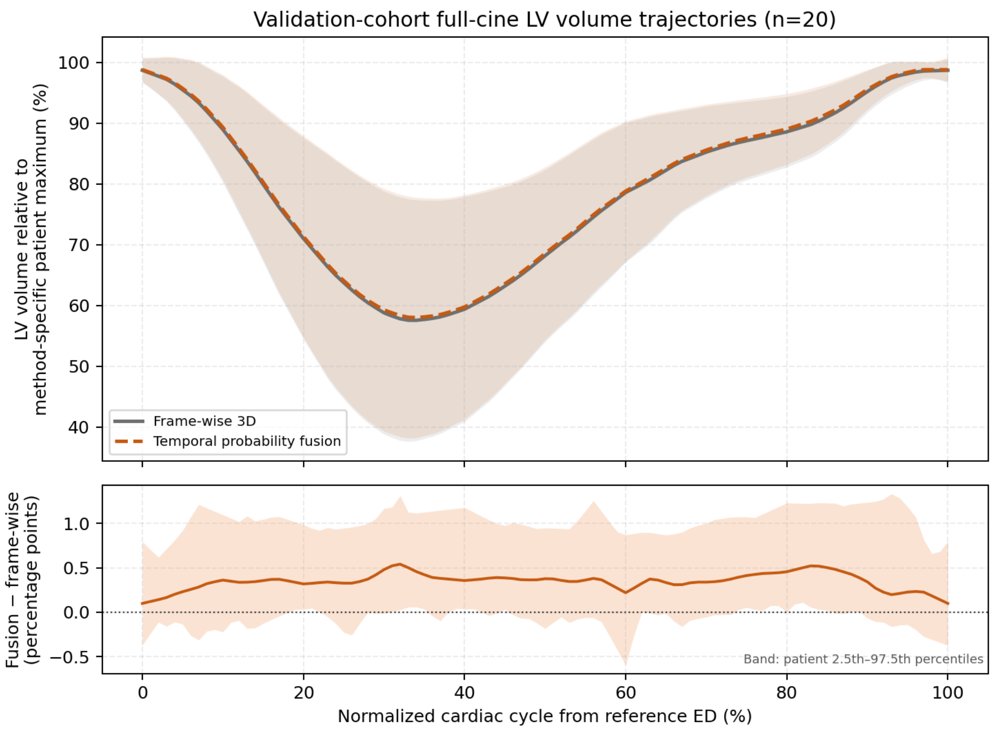


**Figure 6.5: Population LV-volume trajectories for frame-wise inference and fixed temporal fusion across 20 validation patients. The upper bands are mean $\pm$ one between-patient standard deviation; the lower panel shows the paired fusion-minus-frame-wise difference with pointwise patient 2.5th–97.5th percentiles. Each method is normalized by its patient-specific maximum after alignment to reference ED. Temporal fusion is fixed post-hoc probability averaging, not a learned temporal network.**

At the manually annotated ED and ES phases, the frame-wise EF mean absolute error was 3.514 percentage points with Pearson correlation $r = 0.9809$ against mask-derived reference EF. Temporal fusion obtained 3.329 percentage points and $r = 0.9803$. Figure 6.6 shows these patient-level values; the correlation indicates strong linear agreement within this internal cohort but does not constitute independent clinical validation.

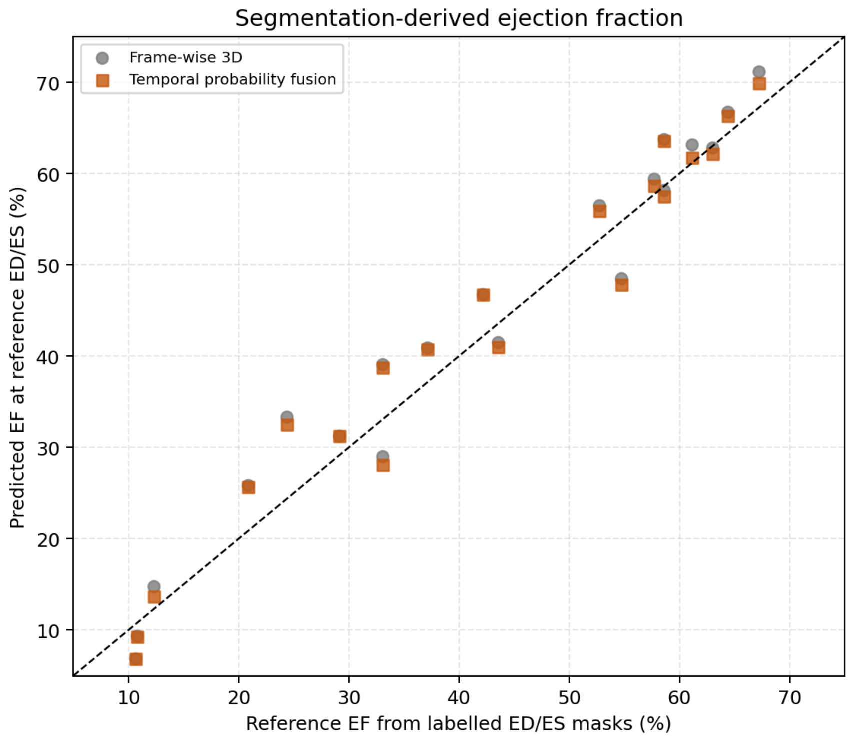


**Figure 6.6: Predicted versus manual-mask-derived EF at the annotated ED and ES phases. Each point is one validation patient. The same cohort was used historically for checkpoint selection, so this is internal validation rather than an independent test.**

### 6.7.3 Phase timing and global motion surrogates

For frame-wise inference, ED was detected exactly for 50% of patients and within one frame for 80%; ES was exact for 55% and within one frame for 85%. Temporal fusion changed the ED prediction for one patient, giving 55% exact while retaining 80% within one frame. ES detection was unchanged. The temporal-fusion median peak LV-centroid displacement from annotated ED was 6.252 mm, with an observed range of 2.152–13.593 mm. Its mean myocardial radial-distance ED-minus-ES change was

4.153 mm. These are global mask-derived surrogates: they summarize position and radial extent, not tissue correspondence or strain.

Patient016 was the main phase-detection outlier for temporal fusion, with ED and ES errors of five and three frames respectively. Its predicted LV curve contains a higher late-cycle maximum than the reference ED region. This failure is retained rather than removed, because it demonstrates that temporal smoothing cannot guarantee correct phase selection when the underlying spatial predictions create a competing volume peak.

### 6.7.4 Qualitative and subgroup observations

Figure 6.7 provides a traceable qualitative example from validation patient002 using the selected epoch-121 Nano-Mamba checkpoint. It compares manual and predicted endpoint masks on the native image grid. The panel illustrates the executed output but is not used as population-level performance evidence.

The diagnostic composition was MINF $n = 7$, NOR $n = 6$, DCM $n = 4$, RV abnormality $n = 2$, and HCM $n = 1$. Temporal-fusion resized endpoint Dice ranged from 82.42% for MINF to 87.56% for NOR, while EF MAE ranged from 2.21 percentage points for DCM to 4.93 for MINF. These values are reported only as descriptive case summaries because the subgroups are small and highly imbalanced; no pathology-level significance or generalization claim is made.

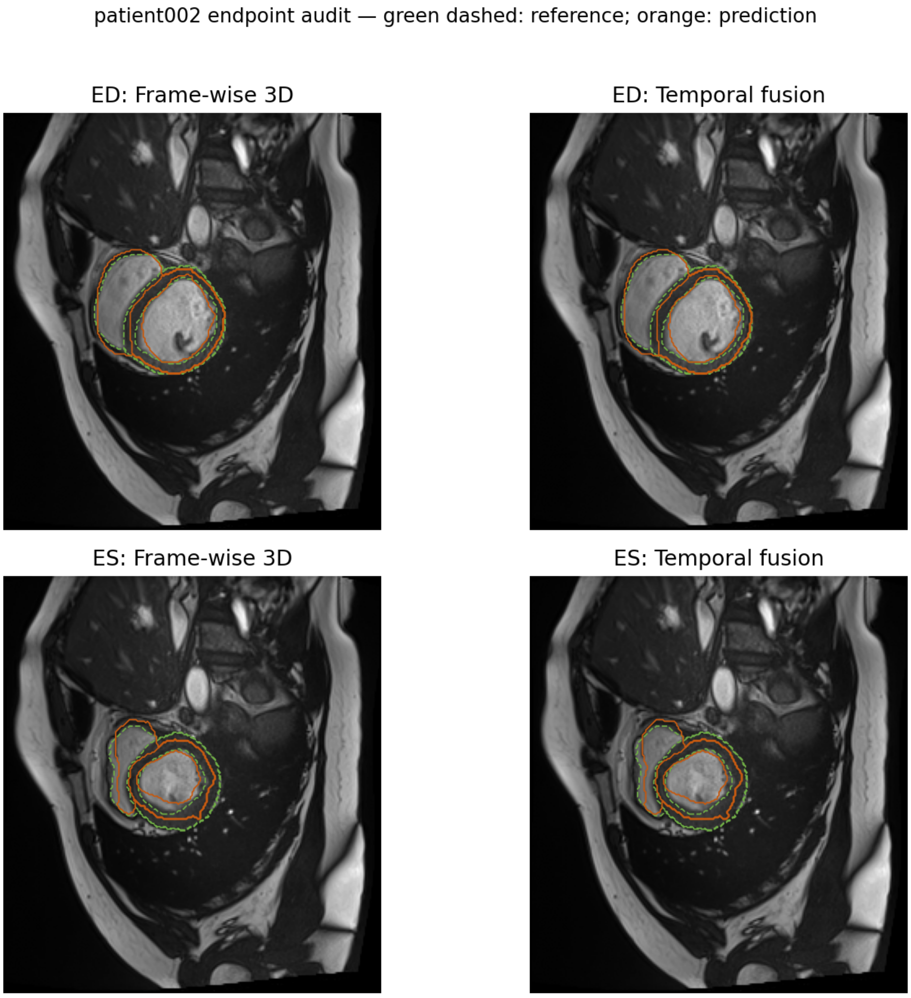


**Figure 6.7: Native-grid ED/ES overlay for validation patient002. Rows show ED and ES; columns show frame-wise and temporal-fusion predictions. Green dashed contours are the manual reference and orange contours are the prediction. This is a single traceable example rather than proof of general performance.**

# CHAPTER 7: DISCUSSION AND LIMITATIONS

This chapter interprets the final validation results and explains the implications for the proposed Nano-Mamba U-Net. The discussion is intentionally conservative: the model is evaluated as a lightweight segmentation architecture, not as an absolute state-of-the-art solution. This distinction is necessary because the final results show a trade-off rather than a universal win.

## 7.1 Accuracy-Efficiency Trade-off

The strongest model by Dice was SegResNet16, with a validation mean DSC of 86.70%. Nano-Mamba U-Net achieved 84.78%, which is 1.92 percentage points lower. Therefore, the final experiment does not support the claim that Nano-Mamba U-Net is the most accurate model overall.

However, Nano-Mamba U-Net achieved this performance using only 1.46M parameters, the smallest parameter count among all evaluated models. Relative to the 3D U-Net baseline, Nano-Mamba improved mean DSC by 3.95 percentage points while using substantially fewer parameters. This supports the central contribution of the thesis: Nano-Mamba U-Net provides a compact architecture with competitive validation performance, suitable for exploring lightweight cardiac segmentation.

The trade-off can be interpreted through the relationship between model size and segmentation accuracy. If $D$ denotes mean Dice and $P$ denotes parameter count, then an informal efficiency ratio can be written as:

$$E = \frac{D}{P}. \tag{7.1}$$

This ratio should not be treated as a clinical metric, but it helps illustrate the engineering perspective. Nano-Mamba U-Net has a lower Dice than SegResNet16 but also a much lower parameter count. Therefore, its value is not that it dominates every metric, but that it occupies a favorable lightweight region of the accuracy-efficiency space.

## 7.2 Interpretation of Ablation Results

The ablation results require careful interpretation. The convolutional-bottleneck control (No-Mamba), which contains zero Mamba operations, achieved 85.64% mean DSC, slightly higher than Nano-Mamba U-Net. This indicates that the current lightweight Mamba-inspired bottleneck does not improve Dice over that stronger convolutional bottleneck on the final validation split. Therefore, the Mamba-inspired module should not be presented as conclusively superior in segmentation accuracy.

At the same time, Nano-Mamba U-Net reduced the parameter count relative to the convolutional control from 2.29M to 1.46M (36.3%) while retaining approximately 99.0% of its Dice and essentially the same recorded FPS (29.09 versus 28.80). Relative to the 64-channel Mamba-bottleneck ablation, Nano-Mamba also used approximately 11.1% fewer reported trainable parameters (1.456M versus 1.638M) while retaining 99.8% of its mean DSC; the raw difference was only $-0.174$ percentage points. Because the paired interval for Nano minus this ablation crosses zero, the experiment does not establish a reliable accuracy penalty for Nano-Mamba. Together, these comparisons support a lightweight accuracy–efficiency interpretation, not a guaranteed accuracy gain from the gate or a formal claim of model equivalence.

A possible explanation is that the compressed bottleneck operates on a very low-resolution representation. At this stage, the network may already have lost some fine anatomical boundary information. A sequence module can mix information across the compressed representation, but it cannot recover details that were not preserved by the encoder. This explains why the Mamba-inspired module may help compactness without guaranteeing the best Dice. Future work should investigate whether placing sequence modules at multiple resolutions would improve this balance.

## 7.3 Class-Specific Observations

LV segmentation was the easiest class across the main models, while MYO segmentation was more difficult. This pattern is clinically reasonable because MYO is a thinner structure and its boundary can be affected by papillary muscles and partial volume effects. RV segmentation also remains challenging because the right ventricle has a more variable shape than the left ventricle, especially in pathological cases.

The class-specific findings support the need for reporting RV, MYO, and LV separately. If only mean Dice were reported, the relatively strong LV score could hide weaker MYO performance. Since cardiac measurements such as myocardial mass depend on MYO segmentation, MYO accuracy cannot be treated as a secondary detail. A future version of the project should therefore include additional metrics such as Hausdorff Distance or Average Surface Distance to better evaluate boundary quality.

## 7.4 Relevance to Cardiac Motion Analysis

The completed project now contains a real full-cine analysis, but its two stages must not be conflated. The trained Nano-Mamba backbone remains a spatial 3D segmentation network: in its input tensor, $D$ denotes anatomical slice depth rather than time, and ED/ES cases were independent during training. Temporal information enters only after inference through a fixed weighted average of adjacent probability maps. The full-cine stage then derives volumes, ED/ES timing, EF, LV-centroid displacement, and myocardial radial-distance change from the resulting mask sequence.

This is a segmentation-derived spatio-temporal framework because it processes every cardiac phase and evaluates how global anatomical quantities vary across the cycle. It is not dense cardiac motion tracking because no optical flow, deformable registration, voxel correspondence, material trajectory, or regional strain field is estimated. The distinction gives a precise answer to the thesis title: complete temporal trajectories and functional indices were produced, but learned temporal representation and tissue-level motion remain future extensions.

The temporal-fusion result also prevents an overly simple claim. Smoother trajectories do not automatically mean more accurate anatomy. Endpoint Dice and EF differences were unresolved, while the smoothness difference was consistent. Fixed fusion is therefore best interpreted as a low-cost temporal regularizer for global trajectories, not as evidence that the model learned cardiac dynamics.

## 7.5 Limitations

Several limitations must be acknowledged.

1. **Internal validation and checkpoint reuse:** The final experiment used a deterministic patient-level validation split of the ACDC training cohort. The same validation patients were inspected across epochs to select each checkpoint and were then used for the descriptive result table. This is not an independent test set and does not represent official ACDC challenge performance.
2. **Single split, seed, and post-hoc uncertainty:** Only seed 42 and one 80/20 patient split were reported. The available per-case CSVs support patient-level bootstrap intervals and paired differences, but these were computed after the experiment and use the same validation patients involved in checkpoint selection. They do not replace multi-seed training or an independent test cohort.
3. **Mamba-inspired implementation:** The implemented bottleneck is inspired by Mamba-style sequence modeling but is not a full reproduction of the original

hardware-aware selective scan implementation. This distinction is important because the thesis must not attribute the full original Mamba mechanism to the implemented model.

4. **Legacy projection parameters:** The historical gate projection produces $2d_{state}+1$ channels but uses only the first. The reported parameter count includes the unused outputs. They were retained to preserve checkpoint compatibility, but a redesigned block would need retraining.
5. **Spatial backbone and fixed temporal processing:** The trained model segments each 3D frame independently; depth is anatomical slices, not time. The complete-cine extension uses fixed adjacent-frame probability fusion and global mask trajectories. It is not a learned temporal Mamba block, optical flow, dense registration, tissue correspondence, or regional strain analysis.
6. **Non-matched training and ablations:** Memory constraints required batch size 1 for Attention U-Net and SegResNet16 and batch size 2 for the other models. Optimizer updates per epoch and normalization choices therefore differ across model families. The zero-Mamba convolutional control also has more parameters and a structurally different bottleneck. These factors limit causal attribution of performance differences to one architectural mechanism.
7. **Array-size preprocessing and no augmentation:** Training resizes arrays to $256 \times 256 \times 16$ without explicit orientation canonicalization, common physical-spacing resampling, cardiac cropping, or augmentation. The historical Dice comparison is therefore measured on a deterministic resized grid. The cine extension restores categorical masks to the native array shape for physical measurements, where endpoint mean Dice is approximately 77.92%, materially lower than the approximately 84.79% resized-grid value. Source review confirms the intended

nearest-neighbour inverse resize with the original three-axis shape and no axis permutation. The gap should nevertheless not be attributed solely to through-plane anisotropy: it combines irreversible boundary discretization from direct array-size resizing, heterogeneous in-plane and through-plane shapes, and model error under a changed evaluation grid. A native-label round-trip audit is provided in the repository to quantify the resampling component on the submission dataset without retraining.

8. **Temporal probability storage precision:** The full-cine run computed logits and softmax in float32 with AMP disabled, preserved the frame-wise argmax before conversion, and stored probability maps as float16 before fixed temporal fusion to reduce host-memory use. The reported fusion result therefore describes that executed mixed-storage pipeline. Because the observed Dice and EF differences are already unresolved, they are not presented as precision-independent improvements; a future sensitivity run should compare float16 and float32 probability storage directly.
9. **Metric convention:** An empty prediction and empty reference for one class are assigned Dice 1 in the executed metric. None of the 720 audited foreground class scores equals 1, establishing that this branch did not affect the reported case table; the convention could still matter in future datasets.
10. **Incomplete temporal ground truth and small subgroups:** ACDC provides manual masks only at ED and ES, so the 510 intermediate-frame masks have no direct overlap ground truth. Diagnostic summaries contain only one to seven patients per group and are descriptive; they cannot establish pathology-specific performance.
11. **Limited external validation:** The model was not evaluated on a separate multi-center external dataset. Generalization to other scanners and institutions remains future work.

12. **Limited spatial metrics and qualitative evidence:** No Hausdorff Distance, Average Surface Distance, probability-calibration metric, regional strain validation, or blinded qualitative review is available. The single patient002 overlay illustrates the validated inference path but does not establish population performance. FPS remains a hardware- and implementation-specific engineering indicator rather than a universal benchmark.
13. **Metadata inconsistency in six recovered cines:** Six public ACDC 4D files have affine metadata that differs from their standalone endpoint container. Shape, header voxel sizes, endpoint pixel content, and endpoint image-label alignment passed the identity gates, and physical metrics use header voxel sizes; nevertheless, a harmonized official archive would be preferable for external reproduction.

These limitations define the thesis boundary. The case rows, training logs, 550-frame table, patient metrics, figures, hashes, and independent recomputation are internally consistent, but they do not remove the design limitations above. Clinical deployment, learned temporal modeling, dense tissue motion, and external generalization remain outside scope.

## 7.6 Summary of Findings

The final validation results position Nano-Mamba U-Net as a lightweight segmentation model with the lowest reported parameter count and competitive recorded batch-one throughput. It outperformed the 3D U-Net baseline in mean DSC and used fewer parameters than all evaluated alternatives, while SegResNet16 and the zero-Mamba convolutional control achieved higher Dice. The added full-cine stage completed segmentation-derived temporal and functional analysis across all 550 frames. Fixed fusion produced a resolved improvement only in global curve smoothness, not in endpoint Dice or EF error. The balanced conclusion is therefore that the project delivers a compact spatial model and an auditable full-cine analysis, without proving the model is the most accurate or that it performs dense cardiac motion tracking.

# CHAPTER 8: CONCLUSION AND FUTURE WORK

This chapter concludes the thesis by summarizing the final experimental findings, answering the research questions, and identifying future directions. The conclusion is based on both the revised patient-level segmentation experiment and the complete-cine post-hoc analysis reported in Chapter 6, together with the interpretation in Chapter 7. The purpose is not to overstate the proposed model, but to identify precisely what was achieved, what was not achieved, and what remains open.

## 8.1 Restatement of the Research Aim

The aim of this thesis was to design and evaluate a lightweight Mamba-inspired U-Net architecture for cardiac MRI segmentation and to determine how its segmentations can support a bounded full-cine cardiac function and motion analysis. The motivation came from a practical research problem: cardiac segmentation models should not only achieve high Dice scores, but should also be computationally manageable, reproducibly evaluated, and connected transparently to downstream cardiac-cycle measurements. Nano-Mamba U-Net integrates a sequence-processing bottleneck into a 3D U-Net-style encoder-decoder network. Compressed volumetric features are serialized into a one-dimensional spatial sequence, processed by a lightweight gated module inspired by Mamba, and reshaped back into the 3D feature space.

This thesis does not claim that Nano-Mamba U-Net is a complete implementation of the original Mamba selective scan algorithm. It also does not claim official ACDC test-set performance, dense tissue motion, regional strain, or clinical deployment readiness. Its contribution is more specific: it implements and evaluates a compact Mamba-inspired bottleneck, reports the resulting accuracy–efficiency trade-off, and applies the selected spatial checkpoint to every phase of the validation cines to derive auditable global motion and functional trajectories.

## 8.2 Summary of Research Contributions

The main contributions are summarized as follows:

1. **Architecture implementation:** A Nano-Mamba U-Net architecture was implemented for 3D cardiac MRI segmentation of RV, MYO, and LV. The model uses a compact encoder-decoder structure and a Mamba-inspired bottleneck at the compressed latent representation. This contribution is architectural integration rather than invention of the original U-Net or Mamba methods.
2. **Rigorous validation protocol:** A deterministic patient-level 80/20 split was used to avoid leakage between training and validation patients. The final experiment used 160 training cases and 40 held-out validation cases from the ACDC training cohort. This corrected the weakness of earlier exploratory scripts that were not sufficiently strict about held-out validation.
3. **Accuracy-efficiency analysis:** Nano-Mamba U-Net achieved 84.78% mean DSC with a post-hoc patient-bootstrap 95% interval of 82.05–86.90% and 1.46M parameters. It improved over the 3D U-Net baseline in Dice while using fewer parameters. However, it did not achieve the highest Dice score overall; SegResNet16 obtained the highest validation mean DSC of 86.70%.
4. **Transparent ablation findings:** The zero-Mamba convolutional-bottleneck control achieved 85.64% mean DSC, slightly higher than Nano-Mamba U-Net. Nano retained approximately 99.0% of that Dice with 36.3% fewer parameters and essentially the same recorded FPS. This supports a lightweight trade-off, not a conclusively superior accuracy mechanism.
5. **Complete-cine analysis:** The selected Nano-Mamba checkpoint was applied to

all 550 phases from 20 validation patients. The workflow reconstructs native-grid masks, LV/RV/MYO volume trajectories, ED/ES timing, EF, LV-centroid displacement, and a global myocardial radial-distance surrogate, while preserving every frame and patient row for audit.

6. **Temporal-fusion finding:** Fixed circular adjacent-frame probability fusion did not establish a resolved endpoint-Dice or EF improvement, but it reduced normalized LV-curve second difference for 19 of 20 patients with a paired 95% bootstrap interval below zero. This distinguishes a supported smoothness result from unsupported learned-motion claims.
7. **Thesis-level scientific correction:** The final thesis deliberately avoids unsupported claims such as absolute state-of-the-art superiority, complete Mamba implementation, learned temporal Mamba behavior, dense cardiac motion tracking, or clinical readiness. A rigorous research project should report both positive and negative evidence accurately.

## 8.3 Answers to Research Questions

**Research Question 1: How can compressed 3D cardiac MRI feature maps be serialized and processed by a Mamba-inspired bottleneck within a U-Net-style segmentation architecture?**

The project answers this question through the Nano-Mamba U-Net design. A 3D encoder first compresses the MRI volume into a lower-resolution latent feature map. This tensor is reshaped from $B \times C \times H \times W \times D$ into a one-dimensional sequence of $HWD$ tokens, processed by a lightweight Mamba-inspired gated block, and then reshaped back into volumetric form before the decoder reconstructs the segmentation mask. The executed block uses kernel-3 depthwise mixing and a token-wise scalar gate; it has no selective scan, recurrent state propagation, attention, or direct global all-token interaction. This demonstrates a practical compact sequence-inspired transformation, not a full Mamba/SSM implementation.

The significance of this design is that the gated processing is applied at the bottleneck, where the spatial resolution is already reduced. This keeps the computational cost manageable. If the same operation were applied directly to the full-resolution input volume, the token length would be much larger and the memory burden would increase substantially. Therefore, the bottleneck placement reflects an engineering compromise between local serialized mixing and computational feasibility.

However, this design also has a limitation. Because the bottleneck operates after several down-sampling steps, some fine boundary information may already have been compressed by the encoder. The sequence module can mix the remaining latent features, but it cannot fully recover details that were lost earlier. This helps explain why the architecture is compact and competitive but not the highest-performing model in the final validation experiment.

**Research Question 2: Does the proposed Nano-Mamba U-Net achieve competitive held-out validation Dice compared with 3D U-Net, Attention U-Net, SegResNet, and ablation variants?**

The answer is partially yes. Nano-Mamba U-Net achieved 84.78% mean DSC, outperforming the 3D U-Net baseline (80.83%) and Attention U-Net (74.78%). The paired post-hoc intervals for Nano minus these two models lie above zero. Nano-Mamba was lower than SegResNet16 (86.70%) and the zero-Mamba convolutional control (85.64%), with paired intervals below zero. The 64-channel Mamba-bottleneck ablation obtained 84.95%, but its paired interval against Nano crosses zero. Therefore, Nano-Mamba U-Net achieved competitive validation performance, but it should not be described as the top-performing model by Dice or as statistically validated on an independent cohort.

This result is scientifically useful because it avoids a simplistic conclusion. If only the comparison against 3D U-Net were considered, Nano-Mamba U-Net would appear clearly better. If only the comparison against SegResNet16 were considered, Nano-Mamba U-Net would appear weaker. The complete interpretation requires considering all models together. The proposed model improves over some baselines, remains close to some ablations, and trails the strongest residual segmentation model.

The final answer to this research question is therefore conditional: Nano-Mamba U-Net is competitive in held-out validation performance, but it is not the strongest model by absolute Dice. This conclusion is more cautious than the original thesis draft, but it is more defensible for academic assessment.

**Research Question 3: What accuracy-efficiency trade-off is observed when comparing Nano-Mamba U-Net with heavier or non-Mamba alternatives?**

Nano-Mamba U-Net achieved the smallest reported parameter count among all evaluated models, with 1.46M parameters. Using the unrounded CSV values, it used approximately 69.7% fewer reported parameters than the 3D U-Net baseline, approximately 69.0% fewer than SegResNet16, 36.3% fewer than the zero-Mamba convolutional control, and 11.1% fewer than the 64-channel Mamba-bottleneck ablation. Against the latter ablation, Nano retained 99.8% of mean DSC and was only 0.174 percentage points lower; the paired interval crossed zero, so no reliable accuracy penalty was established, although formal equivalence was not tested. The reported parameter count includes unused legacy gate-projection outputs, so it describes serialized trainable tensors rather than effective parameters only. The trade-off is that this compactness came with lower Dice than SegResNet16 and the convolutional control. The final interpretation is that Nano-Mamba U-Net offers a lightweight design, but not the highest absolute segmentation accuracy.

This trade-off is central to the thesis. The project should not be judged only by whether it wins a single Dice ranking. A lightweight model can still be meaningful when it provides acceptable accuracy with fewer trainable parameters. At the same time, parameter efficiency alone is not enough; the segmentation quality must remain clinically and technically reasonable. The final result suggests that Nano-Mamba U-Net sits in a useful middle ground: it is smaller than all evaluated models and more accurate than the basic 3D U-Net baseline, but it does not replace SegResNet16 as the accuracy leader.

**Research Question 4: How can the selected spatial segmentation checkpoint support full-cine cardiac function and global motion analysis, and what measurable effect does fixed adjacent-frame probability fusion have relative to frame-wise inference?**

The selected epoch-121 checkpoint was applied independently to all 550 3D phases from the 20 validation cines. Each frame followed the checkpoint-compatible resize and normalization path. Predicted categorical masks were then restored to the native spatial array shape. Header voxel sizes converted class counts and centroids into millilitres and millimetres. LV-volume trajectories supported ED/ES detection, EDV, ESV, stroke volume, and EF; successive masks also supported global centroid-displacement and myocardial radial-distance trajectories. Endpoint image identity, historical Dice reproduction, manual-mask endpoint Dice, EF, phase timing, artifact hashes, and all aggregate arithmetic were audited.

Relative to frame-wise inference, temporal fusion changed resized-grid endpoint Dice from 84.779% to 84.794% and EF MAE from 3.514 to 3.329 percentage points. Both paired bootstrap intervals included zero, so neither change is a resolved improvement. The normalized LV-curve second difference fell from 0.03252 to 0.03139; the paired interval $[-0.00195, -0.00051]$ excluded zero and 19 of 20 patient curves were smoother. Thus, the supported temporal contribution is smoother segmentation-derived global trajectories from a fixed post-hoc filter. It is not evidence that the network learned cardiac time or tissue motion.

## 8.4 Implications of the Findings

The findings have four main implications. First, U-Net-style architectures remain strong foundations for cardiac MRI segmentation. The success of SegResNet16 and the zero-Mamba convolutional control shows that convolutional and residual components are still highly effective on this dataset. This prevents the thesis from making an exaggerated claim that new sequence-inspired modules automatically outperform established convolutional methods.

Second, lightweight sequence-inspired modules remain worth exploring. Nano-Mamba U-Net achieved competitive validation performance with the lowest parameter count. This suggests that compact sequence processing at the bottleneck can be integrated into medical segmentation architectures without causing severe performance collapse. The result is not a final proof of superiority, but it is a reasonable experimental basis for further study.

Third, the validation protocol matters as much as the architecture. The revised patient-level split produced more conservative and more credible results than earlier exploratory evaluations. This is an important lesson for medical AI research: high Dice scores are less meaningful if the evaluation protocol is not strict. A lower but properly validated score is more useful than an inflated score obtained from a weak experimental design.

Fourth, spatial segmentation can support useful complete-cycle measurements without being confused with dense motion tracking. The full-cine stage generated EF, phase, volume, centroid, and radial-distance trajectories, and the fixed fusion improved curve smoothness. At the same time, the unresolved Dice/EF changes and the absence of intermediate manual masks show why every downstream quantity still needs its own validation boundary.

## 8.5 Future Work

Several directions can strengthen this work further.

### 8.5.1 Evaluation on External and Official Test Sets

Future experiments should evaluate the model on the official ACDC test set or on independent external cardiac MRI datasets. This would provide stronger evidence of generalization beyond the internal patient-level validation split. External validation is particularly important in medical imaging because scanner vendor, acquisition protocol, image resolution, and patient population can all affect model performance.

### 8.5.2 Larger Pathology-Specific Validation

The completed cine analysis reports descriptive diagnostic-group summaries from actual ACDC metadata, but the validation cohort contains only one HCM patient and two abnormal-RV patients. Future work should use larger, externally sampled groups and pre-specified subgroup analyses before drawing pathology-specific conclusions. This would allow a clinically more meaningful study of where segmentation and EF estimation fail without over-interpreting very small internal groups.

### 8.5.3 Additional Boundary-Based Metrics

The current thesis focuses mainly on Dice Similarity Coefficient. Dice is suitable for overlap-based evaluation, but it does not fully describe boundary quality. Future work should report Hausdorff Distance, Average Surface Distance, or related boundary metrics. This is especially relevant for myocardium segmentation, where small boundary errors can affect clinical measurements even when overlap scores appear acceptable.

### 8.5.4 Closer Integration with Full Mamba/SSM Implementations

The current implementation is Mamba-inspired rather than a full selective scan implementation. Future work could compare the lightweight PyTorch block against an official or optimized Mamba implementation to determine whether a more faithful state space module improves cardiac segmentation. Such a comparison would clarify whether the current result is limited by the simplified bottleneck or by the broader suitability of sequence modeling for compressed cardiac features.

### 8.5.5 Multi-Scale Sequence Modeling

The present model applies the Mamba-inspired module only at the bottleneck. Future work could place rigorously specified sequence or state-space modules at multiple encoder or decoder resolutions. This may help preserve finer boundary information while enabling richer latent interactions. However, multi-scale sequence modeling would also increase memory usage and implementation complexity, so it should be evaluated carefully rather than assumed to be superior.

### 8.5.6 Learned Temporal and Dense Motion Modeling

The present study already processes complete 4D cine sequences and derives global segmentation-based trajectories, but the trained network remains spatial and the temporal fusion is fixed. A stronger extension would jointly input multiple phases and learn temporal representations under explicit consistency supervision. Such a model should be compared against the current frame-wise and fixed-fusion baselines, rather than assuming that a learned temporal block is automatically better.

Dense motion modeling is a separate extension. Optical flow or deformable registration could establish spatial correspondence between myocardial points, enabling regional displacement and strain rather than only global centroid and radial-distance surrogates.

This would require suitable ground truth or independently validated clinical measurements, topology checks, and evaluation beyond ED/ES overlap.

## 8.6 Concluding Remarks

This thesis demonstrates that a compact Mamba-inspired bottleneck can be integrated into a 3D U-Net-style architecture for cardiac MRI segmentation. The final segmentation experiment does not show absolute state-of-the-art accuracy, but it does show competitive internal-validation performance with the smallest parameter count among the evaluated models. The complete-cine extension further demonstrates how one frozen spatial checkpoint can generate auditable cardiac-cycle volume, function, phase, and global motion trajectories under practical hardware constraints.

The most important conclusion is that the proposed model should be understood as an accuracy–efficiency trade-off model embedded in a completed but bounded spatio-temporal workflow. It is not a universal replacement for stronger residual segmentation networks and not a complete clinical motion-analysis system. The evidence supports the internal six-model comparison, full-cine segmentation-derived functional analysis, and a curve-smoothness benefit from fixed temporal fusion. Independent testing, multi-seed and matched-capacity training, native-space model training, boundary metrics, larger subgroup evaluation, learned temporal representations, and dense motion or strain validation would extend this work into a stronger cardiac image-analysis framework.

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